

Hopf Bifurcations and Bistability in the Hodgkin-Huxley Equations

at the 1952 Parameters: Computer-Assisted Proofs

Chase Hendrick

Independent Researcher

chase@hendrickresearch.com

ORCID 0009-0002-9754-6087

Abstract

The space-clamped Hodgkin–Huxley equations with Hodgkin and Huxley’s constants are the standard model of the action potential, and their bifurcations are known from numerical computation. We prove several of them with computer assistance in ball arithmetic, for every leak reversal potential E_l in $[10.59, 10.62]$ mV, an interval that contains the value 10.613 printed by Hodgkin and Huxley and the value $E_l^* = 10.5989209693916 \dots$ that makes the resting current zero, as their Table 3 says it should. For every applied current J in $[0, 200]$ $\mu\text{A}/\text{cm}^2$ there is exactly one equilibrium. There are two Hopf points $J_{H1} < J_{H2}$ in this range: the equilibrium is asymptotically stable for $J < J_{H1}$ and for $J > J_{H2}$, it has exactly two eigenvalues with positive real part for $J_{H1} < J < J_{H2}$, and no eigenvalue lies on the imaginary axis at any current of the range other than J_{H1} and J_{H2} . At $E_l = 10.613$ the Hopf points lie at $J_{H1} \in [9.775437995393, 9.775437995394]$ and $J_{H2} \in [154.522433665808, 154.522433665809]$, and both move by exactly 0.3 ($10.613 - E_l$) when E_l changes. The eigenvalues cross the imaginary axis transversally, and the first Lyapunov coefficient is positive at J_{H1} and negative at J_{H2} : the lower Hopf bifurcation is subcritical and the upper one supercritical. At $J = 8$, for every E_l in the interval, a locally asymptotically stable equilibrium coexists with an orbitally asymptotically stable periodic orbit, a train of action potentials with period between 16.0058 and 16.0140 ms: the model is bistable there. At three values of E_l we also enclose a periodic orbit of saddle type. The proofs combine a Routh–Hurwitz analysis along the branch of equilibria with a C^1 Lohner-type Taylor integrator, the Krawczyk test and a contraction argument on Poincaré sections. Nothing is claimed about the basins of attraction or about other attractors.

Keywords: Hodgkin–Huxley equations; Hopf bifurcation; first Lyapunov coefficient; bistability; periodic orbit; computer-assisted proof; ball arithmetic.

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1 Introduction

Hodgkin and Huxley’s equations for the squid giant axon [19] are the prototype of a conductance-based neuron model. With the membrane potential uniform in space (the space clamp), they form a system of four ordinary differential equations in the depolarization u and three gating variables m , n , h , with an applied current J as the natural parameter. The picture of their dynamics that textbooks and later models rely on is numerical: a single equilibrium, stable for small and large currents and unstable in between; a Hopf bifurcation at each end of the unstable range, subcritical at the lower end near $J \approx 9.78 \mu\text{A}/\text{cm}^2$; and, below that current, a range in which the resting state coexists with repetitive firing, so that a brief pulse can start or stop a train of action potentials. Lu,

Xin and Rinzel [31, pp. 61–62] review how the coexistence was found in simulations (they credit Cooley, Dodge and Cohen) and observed in the squid axon by Guttman, Lewis and Rinzel [14]; Best [2] showed numerically that a suitable brief pulse stops the firing for good. This paper proves parts of that picture at Hodgkin and Huxley’s own constants.

What is known. The bifurcations of the space-clamped equations have been computed numerically many times; we cite only what we have read, and Section 10 says how far. Troy proved a Hopf bifurcation for a modified system in which n and h are slowed by a small parameter, under assumptions that he describes as “made on the basis of numerical evidence and physical reasoning” [38, p. 76], and large-amplitude periodic solutions for a reduced system of three equations under assumptions [37]; his earlier paper [36] proves Hopf bifurcation “under appropriate assumptions” (abstract). None of these is a proof for the full system at the 1952 constants. Hastings writes that the parameters “appear, on the basis of numerical calculations, to be such that the system (3)–(4) has a unique equilibrium point” [18, p. 230]; his system (3)–(4) is the traveling-wave system, whose equilibria are those of the space-clamped system at zero applied current. Guckenheimer and Labouriau state that when the potassium reversal potential has Hodgkin and Huxley’s value, “ f is monotonic and HH has a unique equilibrium for each value of I ” [11, p. 941], f being the steady-state current (the same sentence is on p. 23 of the arXiv preprint of [13]); they give no proof, their computations of equilibria and of local bifurcations are symbolic (Macsyma, Mathematica and Maple) and their global ones numerical, and they write that their bifurcation diagrams in the plane of the current and the potassium reversal potential “are conjectured on the basis of symbolic calculations, numerical computations and visual examination of a large number of phase portraits” [11, p. 938] and “have not been proved to be correct” [11, p. 942]. Labouriau showed with singularity theory that the two branches of periodic solutions born at the Hopf points join, for the equations perturbed in parameters such as the temperature, and computed the invariants of the degenerate Hopf bifurcation that organizes them, which give a topological description of the periodic orbits near the equilibrium [28, 29]. Her thesis [27], whose Chapter IV (“Nerve impulse equations”) we have read, computes the eigenvalues along the branch of equilibria, the two Hopf points and the direction of bifurcation in floating point, with $V_L = 10.599$, and takes the invertibility of the steady-state current from a graph: “within physiologically meaningful range f is monotonically increasing (Fig. IV-0-1)”; it states that “all the information we have about HH, after reduction, is a numerical estimate of its derivatives”. Hassard and Shiau computed isolated branches of periodic solutions near degenerate Hopf points, with the temperature and the sodium conductance or reversal potential varied, by “purely numerical techniques” for the global branches [16, 35], and computed, in the same way, a curve of doubly degenerate Hopf points in the parameters (I, T, g_{Na}, v_{Na}) with a special point of Z_2 -codimension three, adding that “we have not solved the unfolding problem” [17, p. 33]. Of [28, 29, 16, 35] we have read the abstracts, and [17] in full. Fukai, Doi, Nomura and Sato computed the bistable periodic solutions in several parameters [8, 9]. Guckenheimer and Oliva continued the periodic orbits born at the subcritical Hopf point and found chaotic dynamics near $J = 7.86$, adding that “we do not give a rigorous proof that chaos exists in this system” [12, p. 106]. The papers cited above also report the numerical Hopf computations of Hassard [15] and the stable and unstable periodic solutions of Rinzel and Miller [33]. Du and Hassard [7] describe “an algorithm . . . for the computation of Hopf bifurcation coefficients for autonomous ordinary differential systems, by Lyapunov–Schmidt reduction”, in which “interval values for the coefficients are found and it is known with mathematical certainty that the exact values fall in the computed intervals”, to solve the location and recognition problems of degenerate Hopf points of Z_2 -codimension at most three, with “applications to an enzyme-catalyzed reaction

model and to the Hodgkin–Huxley nerve conduction model” (abstract); the review in zbMATH adds that nondegeneracy conditions and the genericity of the unfolding are checked. Their method is local, concerned with the equilibrium and the periodic solutions near a Hopf point, and interval enclosures of the Hopf points of the model and of their coefficients may already be found there; we claim no priority for them. Computer-assisted proofs exist for the traveling pulse of the FitzHugh–Nagumo equations [1], for their periodic traveling waves [5], and for Hopf bubbles and degenerate Hopf bifurcations [4]; validated continuation of periodic orbits from a Hopf bifurcation is a general method [3].

This paper. We prove, for every leak reversal potential $E_l \in [10.59, 10.62]$:

- Theorem 1: for every $J \in [0, 200]$ there is exactly one equilibrium.
- Theorem 2: there are two currents $J_{H1} < J_{H2}$ such that along this branch the equilibrium is asymptotically stable for $J < J_{H1}$ and for $J > J_{H2}$, and has exactly two eigenvalues in $\text{Re } \lambda > 0$ for $J_{H1} < J < J_{H2}$; at J_{Hi} it has a simple pair $\pm i\omega_i$ that crosses the imaginary axis transversally, and the first Lyapunov coefficient ℓ_1 is positive at J_{H1} and negative at J_{H2} . The currents are enclosed to 10^{-12} .
- Corollary 3: the Hopf bifurcation at J_{H1} is subcritical and the one at J_{H2} supercritical, in the sense of the normal form.
- Theorem 4: at $J = 8$ there is an orbitally asymptotically stable periodic orbit, a spike train with period in $[16.005827509, 16.013912063]$ ms.
- Theorem 5: at $J = 8$ and $E_l = 10.613$, E_l^* and 10.599 , a stable and a saddle-type periodic orbit, with narrow enclosures of their periods and Floquet multipliers; the stable one is the orbit of Theorem 4 (Corollary 7).
- Corollary 6: bistability of rest and repetitive firing at $J = 8$, equivalently for $E_l = 10.613$ and every $J \in [7.9931, 8.0021]$.

We have not found an earlier proof that the space-clamped Hodgkin–Huxley equations at the 1952 constants have a stable periodic orbit of large amplitude away from the Hopf points, such as the spike train of Theorem 4, or that rest and repetitive firing are bistable. We claim no priority for Theorems 1 and 2 or Corollary 3: Guckenheimer and Labouriau state the uniqueness of the equilibrium without proof, Du and Hassard’s interval computations may already prove statements about the branch of equilibria, locate the Hopf points and fix their criticality, and a proved $\ell_1 < 0$ at J_{H2} gives, with the Hopf theorem, the small stable cycles of Corollary 3(b). Du and Hassard describe a local method, the periodic solutions near a Hopf point, and neither their abstract nor the review in zbMATH mentions a periodic orbit away from the Hopf points or bistability. These statements are limited by the searches behind them, which Section 10 lists with the dates, or the range of dates, on which they were made, and how far each relevant paper was read.

Labels. Every result carries a label. *Proved*: a written proof in this paper, with no computation. *Computer-assisted*: a written proof in which finitely many inequalities are decided by a program in ball arithmetic (Section 7); the program and the lines of its output that carry each step are named. *Cited*: a published theorem used as stated in its source, with its hypotheses checked here; the two cited theorems are lettered, Theorems H and K. A result whose proof is computer-assisted and also uses a cited theorem says so in its label. *Numerical*: floating-point or high-precision computation

without error control; numerical results are never used in a proof and are collected in Section 8. Paths such as `code/certlib.py` name files of the folder that accompanies this paper. Section 9 says how to rerun everything, and Section 10 lists what is not proved and which checks of this work have been made.

2 The model

2.1 The equations

We use the modern sign convention: $u = -V$ is the depolarization from rest in mV, where V is the displacement of Hodgkin and Huxley, and $J = -I$ is the applied depolarizing current in $\mu\text{A}/\text{cm}^2$; time is in ms and the temperature is 6.3°C , so no temperature factor enters. With the rate constants of [19, eqs. (12), (13), (20), (21), (23), (24)], the current equation [19, eq. (26)] and the constants of [19, Table 3] (the membrane capacity is $1 \mu\text{F}/\text{cm}^2$), the space-clamped equations read

$$\begin{aligned} \dot{u} &= J - I(u, m, n, h), & I &= g_{\text{Na}} m^3 h (u - E_{\text{Na}}) + g_{\text{K}} n^4 (u - E_{\text{K}}) + g_l (u - E_l), \\ \dot{y} &= \alpha_y(u)(1 - y) - \beta_y(u)y, & y &\in \{m, n, h\}, \end{aligned} \quad (1)$$

with $g_{\text{Na}} = 120$, $g_{\text{K}} = 36$, $g_l = 0.3$, $E_{\text{Na}} = 115$, $E_{\text{K}} = -12$ and

$$\begin{aligned} \alpha_m &= \Psi\left(\frac{25-u}{10}\right), & \beta_m &= 4e^{-u/18}, & \alpha_n &= 0.1 \Psi\left(\frac{10-u}{10}\right), & \beta_n &= 0.125 e^{-u/80}, \\ \alpha_h &= 0.07 e^{-u/20}, & \beta_h &= \frac{1}{e^{(30-u)/10} + 1}, & \Psi(x) &= \frac{x}{e^x - 1}, & \Psi(0) &= 1. \end{aligned} \quad (2)$$

We write $x = (u, m, n, h)$, $f(x; J, E_l)$ for the right-hand side of (1), $k_y = \alpha_y + \beta_y$ and $y_\infty = \alpha_y/k_y$. Table 3 of [19] prints the leak potential $V_l = -10.613$ mV, that is $E_l = 10.613$, with the footnote “Exact value chosen to make the total ionic current zero at the resting potential ($V = 0$)”.

Lemma 2.1 (proved). (a) $\Psi(x) > 0$ and $\Psi'(x) < 0$ for every real x , so the six rates are positive and $y_\infty(u) \in (0, 1)$ for every $u \in \mathbb{R}$. (b) n_∞ is strictly increasing on $\mathbb{R}$. (c) f is real-analytic on $\mathbb{R}^4 \times \mathbb{R}^2$ (in x , J and E_l). (d) The equilibria of (1) are exactly the points $x^*(u) = (u, m_\infty(u), n_\infty(u), h_\infty(u))$ with $J_{\text{ss}}(u) = J$, where $J_{\text{ss}}(u) = I(x^*(u))$ is the steady-state current.

Proof. (a) x and $e^x - 1$ have the same sign, so $\Psi > 0$. $\Psi'(x) = g(x)/(e^x - 1)^2$ with $g(x) = e^x - 1 - xe^x$ for $x \neq 0$; $g(0) = 0$ and $g'(x) = -xe^x$, so $g < 0$ for $x \neq 0$, and $\Psi'(0) = -1/2$ from the series $\Psi(x) = 1 - x/2 + \dots$. The other rates are exponentials or reciprocals of $e^{(30-u)/10} + 1 > 1$. (b) $n_\infty = 1/(1 + \beta_n/\alpha_n)$; β_n decreases in u , and α_n increases because Ψ decreases and $(10 - u)/10$ decreases. (c) Ψ is analytic on the strip $|\text{Im } x| < 2\pi$ (its only singularities are at $x = 2\pi ik$, $k \neq 0$, since the zero at $x = 0$ of $e^x - 1$ is removable), $1/(e^{(30-u)/10} + 1)$ is analytic near the real axis, and f is polynomial in m , n , h , J , E_l . (d) $\dot{y} = 0$ means $y = \alpha_y/k_y$ since $k_y > 0$; then $\dot{u} = 0$ means $J = J_{\text{ss}}(u)$. $\square$

2.2 The leak potential

Lemma 2.2 (proved). $f(x; J, E_l) = f(x; J + g_l(E_l - E'), E')$ for all x , J , E_l , E' . Hence a statement about the system at (J, E_l) is a statement about the system at $(J + 0.3(E_l - E'), E')$. In particular every bifurcation current $J_*(E_l)$ satisfies $J_*(E_l) = J_*(10.613) + 0.3(10.613 - E_l)$, and the system at $J = 8$ with $E_l \in [10.59, 10.62]$ is the system at $E_l = 10.613$ with $J \in [7.9931, 8.0021]$.

Proof. E_l enters f only through $J + g_l E_l$ in $\dot{u} = (J + g_l E_l) - g_{\text{Na}} m^3 h (u - E_{\text{Na}}) - g_{\text{K}} n^4 (u - E_{\text{K}}) - g_l u$. The last sentence is $8 + 0.3(E_l - 10.613) \in [8 - 0.0069, 8 + 0.0021]$. $\square$

Proposition 2.3 (computer-assisted). *The value of E_l for which the ionic current vanishes at the resting potential $u = 0$, that is $I(x^*(0)) = 0$, is*

$$E_l^* = \frac{-g_{\text{Na}} E_{\text{Na}} m_{\infty}(0)^3 h_{\infty}(0) - g_{\text{K}} E_{\text{K}} n_{\infty}(0)^4}{g_l},$$

and $E_l^* \in [10.5989209693916785221988785, 10.5989209693916785221988786]$. With $E_l = 10.613$ the ionic current at $u = 0$ is $0.3(E_l^* - 10.613) \in [-0.004223709183, -0.004223709182] \mu\text{A}/\text{cm}^2$, not zero.

The value 10.599 used by Fukai et al. [8, p. 216], who list it among the parameter values used by Hodgkin and Huxley, by Guckenheimer and Labouriau [11, p. 939] and Guckenheimer and Worfolk (p. 23 of the arXiv preprint of [13]), who list it among what they call the HH values (printed as $\tilde{V}_L = 10.599$ mV beside $\tilde{V}_{\text{Na}} = -115$ mV, although in that sign convention Table 3 prints $V_l = -10.613$), and by Guckenheimer and Oliva [12] and Labouriau [27, Ch. IV] agrees with E_l^* to the digits given; Hassard and Shiau define the leak potential by the same condition, “adjusted so for $v = 0$ ” the ionic current vanishes at the steady state [17, p. 34]. We have not found the difference between E_l^* and the printed 10.613 discussed in what we read; nothing in this paper depends on it. Because the qualitative statements below hold on the whole interval $E_l \in [10.59, 10.62]$, which contains 10.613, E_l^* and 10.599, the choice between these values does not matter for them, and Lemma 2.2 converts every current from one value to another.

2.3 The Jacobian at an equilibrium

Lemma 2.4 (proved). *At an equilibrium $x^*(u)$ the Jacobian of f is the arrow matrix*

$$A(u) = \begin{pmatrix} a_{11} & a_{1m} & a_{1n} & a_{1h} \\ a_{m1} & -k_m & 0 & 0 \\ a_{n1} & 0 & -k_n & 0 \\ a_{h1} & 0 & 0 & -k_h \end{pmatrix},$$

with $a_{11} = -(g_{\text{Na}} m^3 h + g_{\text{K}} n^4 + g_l)$, $a_{1m} = -3g_{\text{Na}} m^2 h(u - E_{\text{Na}})$, $a_{1n} = -4g_{\text{K}} n^3(u - E_{\text{K}})$, $a_{1h} = -g_{\text{Na}} m^3(u - E_{\text{Na}})$ and $a_{y1} = \alpha'_y(1 - y) - \beta'_y y = k_y y'_{\infty}(u)$, everything at $x^*(u)$. It does not depend on J or E_l . Put $c_y = a_{1y} a_{y1}$, $e_1 = k_m + k_n + k_h$, $e_2 = k_m k_n + k_m k_h + k_n k_h$, $e_3 = k_m k_n k_h$. Then

$$(a) \det(\lambda I - A) = p(\lambda) = \lambda^4 + a_1 \lambda^3 + a_2 \lambda^2 + a_3 \lambda + a_4 \text{ with } a_1 = e_1 - a_{11}, a_2 = e_2 - a_{11} e_1 - \sum_y c_y, \\ a_3 = e_3 - a_{11} e_2 - \sum_y c_y \sum_{y' \neq y} k_{y'}, \text{ and } a_4 = -a_{11} e_3 - \sum_y c_y \prod_{y' \neq y} k_{y'};$$

$$(b) a_4 = e_3 J'_{\text{ss}}(u);$$

$$(c) \text{ if } p(\lambda) = 0 \text{ and } \lambda \notin \{-k_m, -k_n, -k_h\}, \text{ then}$$

$$q = \left(1, \frac{a_{m1}}{\lambda + k_m}, \frac{a_{n1}}{\lambda + k_n}, \frac{a_{h1}}{\lambda + k_h}\right), \quad w = \left(1, \frac{a_{1m}}{\lambda + k_m}, \frac{a_{1n}}{\lambda + k_n}, \frac{a_{1h}}{\lambda + k_h}\right)$$

satisfy $Aq = \lambda q$ and $w^T A = \lambda w^T$.

Proof. The entries follow by differentiating (1); $a_{y1} = \alpha'_y \beta_y / k_y - \beta'_y \alpha_y / k_y = k_y y'_{\infty}$. For $\lambda \notin \{-k_y\}$ the Schur complement of the diagonal block gives $\det(\lambda I - A) = \prod_y (\lambda + k_y) (\lambda - a_{11} - \sum_y c_y / (\lambda + k_y))$, a polynomial identity that holds for all λ ; expanding it gives (a). For (b), $J'_{\text{ss}} = \partial_u I + \sum_y \partial_y I y'_{\infty} = -a_{11} - \sum_y a_{1y} a_{y1} / k_y$, and multiplying by e_3 gives the expression for a_4 . For (c), rows 2 to 4 of $(A - \lambda)q = 0$ read $a_{y1} - (k_y + \lambda)q_y = 0$, and row 1 reads $a_{11} + \sum_y c_y / (\lambda + k_y) - \lambda = 0$, which is $p(\lambda) / \prod_y (\lambda + k_y) = 0$; w is checked in the same way. $\square$

Below, $D_2 = a_1 a_2 - a_3$ and $D_3 = a_1 a_2 a_3 - a_3^2 - a_1^2 a_4 = a_3 D_2 - a_1^2 a_4$ are the Hurwitz determinants of p [21]; like the a_k , they are functions of u alone.

2.4 Periodic orbits

Let φ_t be the flow of (1) for fixed (J, E_l) and Γ a periodic orbit of minimal period T . The *Floquet multipliers* of Γ are the eigenvalues of the monodromy matrix $D\varphi_T(y)$, $y \in \Gamma$, which do not depend on y up to similarity; $f(y)$ is an eigenvector for the multiplier 1. Let $\Sigma = \{x_i = c\}$ be a hyperplane with $f_i(y) > 0$ at $y \in \Gamma \cap \Sigma$. The implicit function theorem, applied to $x_i(\varphi_t(x)) - c = 0$ at $(t, x) = (T, y)$, gives a neighbourhood U of y in Σ and a real-analytic function τ_{oc} on U with $\tau_{\text{oc}}(y) = T$ and $\varphi_{\tau_{\text{oc}}(z)}(z) \in \Sigma$; we call $P(z) = \varphi_{\tau_{\text{oc}}(z)}(z)$ the *local return map* at y . It fixes y , and the *nontrivial* multipliers are the eigenvalues of $DP(y)$; by Lemma 5.8 they are the multipliers other than one copy of 1. Near its fixed points, the first-return map that the programs compute is this map (Lemma 5.7). Γ is *orbitally asymptotically stable* if for every $\varepsilon > 0$ there is $\delta > 0$ such that $\text{dist}(x, \Gamma) < \delta$ implies $\text{dist}(\varphi_t(x), \Gamma) < \varepsilon$ for all $t \geq 0$ and $\text{dist}(\varphi_t(x), \Gamma) \rightarrow 0$; it is *not orbitally stable* if the first property fails. We call Γ of *saddle type* if one nontrivial multiplier is real and larger than 1 and the others have modulus less than 1. The sections used are $\Sigma_{20} = \{u = 20\}$ and $\Sigma_5 = \{u = 5\}$, crossed with $\dot{u} > 0$, with coordinates (m, n, h) .

3 Results

Theorem 1 (computer-assisted). *For every $E_l \in [10.59, 10.62]$ and every $J \in [0, 200]$ the system (1) has exactly one equilibrium $x^*(u^*(J))$, and $-12 < u^*(J) < 115$. The function $J \mapsto u^*(J)$ is real-analytic and strictly increasing.*

Theorem 2 (computer-assisted). *Let $E_l \in [10.59, 10.62]$. The coefficients a_1, a_3, a_4 of Lemma 2.4 are positive on $[-12, 115]$, and D_3 has exactly two zeros there, both simple, $u_{H1} \in [5.345856397004, 5.345856397005]$ and $u_{H2} \in [21.941907987016, 21.941907987017]$, with $D_3 > 0$ on $[-12, u_{H1}) \cup (u_{H2}, 115]$ and $D_3 < 0$ on (u_{H1}, u_{H2}) . Put $J_{Hi} = J_{\text{ss}}(u_{Hi})$. Then:*

- (a) *For $J \in [0, J_{H1}) \cup (J_{H2}, 200]$ every eigenvalue of the equilibrium has negative real part, so the equilibrium is locally asymptotically stable. For $J \in (J_{H1}, J_{H2})$ exactly two eigenvalues, counted with multiplicity, have positive real part, none lies on the imaginary axis, and the equilibrium is unstable.*
- (b) *At $J = J_{Hi}$ the eigenvalues are a simple pair $\pm i\omega_i$ and two eigenvalues with negative real part, with $\omega_1 \in [0.586233813130, 0.586233813131]$ and $\omega_2 \in [1.062921806760, 1.062921806761]$ ms^{-1} ($2\pi/\omega_1 \in [10.717882807, 10.717882808]$ ms , $2\pi/\omega_2 \in [5.911239441, 5.911239442]$ ms).*
- (c) *$J_{Hi}(E_l) = J_{Hi}(10.613) + 0.3(10.613 - E_l)$, with the values $(\mu A/\text{cm}^2)$*

E_l	J_{H1}	J_{H2}
10.613	[9.775437995393, 9.775437995394]	[154.522433665808, 154.522433665809]
E_l^*	[9.779661704575, 9.779661704576]	[154.526657374990, 154.526657374991]
10.599	[9.779637995393, 9.779637995394]	[154.526633665808, 154.526633665809]
[10.59, 10.62]	[9.773337995, 9.782337996]	[154.520333665, 154.529333666]

- (d) *The eigenvalue $\lambda(J)$ with $\lambda(J_{Hi}) = i\omega_i$ crosses the imaginary axis transversally: $\text{Re } \lambda'(J_{H1}) \in [0.01878905889, 0.01878905890]$ and $\text{Re } \lambda'(J_{H2}) \in [-0.00449227917, -0.00449227916]$.*
- (e) *The first Lyapunov coefficient (3), with the eigenvector q normalized by $\langle q, q \rangle = 1$ in the coordinates (u, m, n, h) (u in mV , t in ms), is $\ell_1(H1) \in [0.01479364813100, 0.01479364813101] > 0$ and $\ell_1(H2) \in [-0.004728502695074, -0.004728502695073] < 0$.*

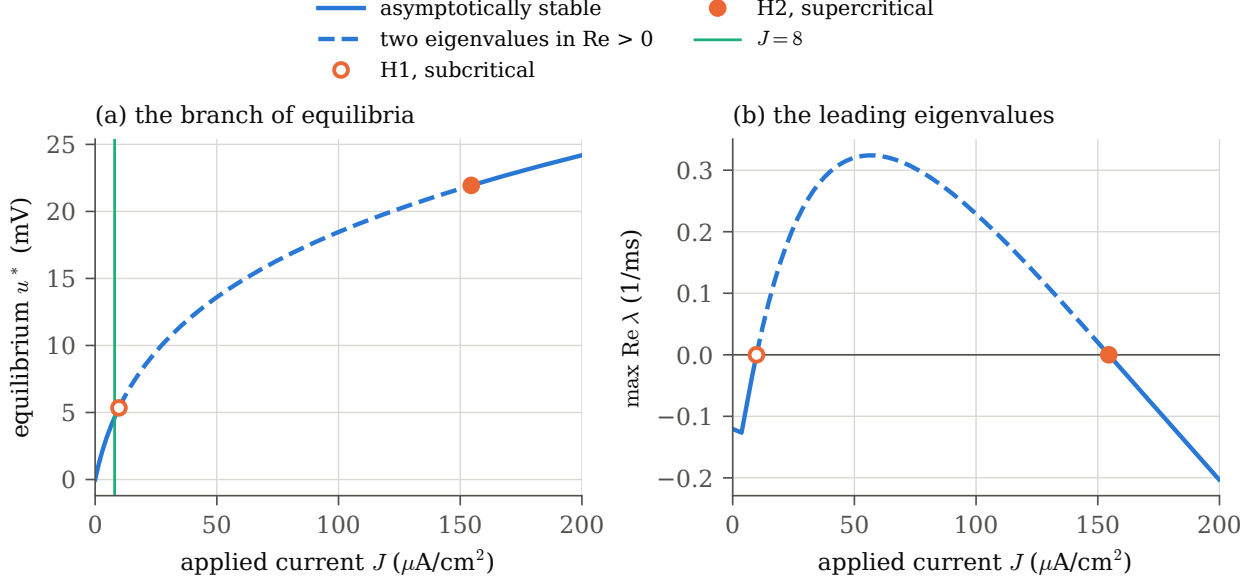


Figure 1: The branch of equilibria at $E_l = 10.613$ (code/hh_make_figures.py). (a) $u^*(J)$ for $J \in [0, 200]$, solid where the equilibrium is asymptotically stable and dashed where it has two eigenvalues with positive real part; circles: the Hopf points of Theorem 2; the vertical line is $J = 8$, the current of Theorems 4 and 5. (b) The largest real part of an eigenvalue. The curves are floating-point evaluations; the positions of the Hopf points and the stability types are those proved in Theorem 2.

The value of ℓ_1 depends on the normalization of q ; its sign does not. Figure 1 shows the branch.

Remark 3.1 (computer-assisted). The checks behind Theorems 1 and 2 prove more: $J_{ss}(u) \geq 4089.4815$ for every $u \geq 115$, and $J'_{ss} > 0$ and the signs of a_1 , a_3 , a_4 and D_3 are established on all of $[-12, 115]$ (Section 4). So for every $E_l \in [10.59, 10.62]$ and every $J \in [0, 4089.4815]$ there is exactly one equilibrium, with $u^*(J) \in (-12, 115)$, and it is asymptotically stable for every $J \in (J_{H2}, 4089.4815)$. We state the theorems on $[0, 200]$, which contains the currents of physiological interest.

Corollary 3 (computer-assisted, with the cited Theorem H). *Let $E_l \in [10.59, 10.62]$. Near $(x^*(J_{Hi}), J_{Hi})$ the one-parameter family (1), with parameter J , is locally topologically equivalent to the suspension of the Hopf normal form by a two-dimensional stable node, with $\sigma = +1$ at J_{H1} and $\sigma = -1$ at J_{H2} (Theorem H). Consequently, in the neighbourhood where the equivalence holds:*

- (a) (subcritical) for $J < J_{H1}$ close to J_{H1} , where the equilibrium is stable, there is a limit cycle, and it is not orbitally stable; for $J \geq J_{H1}$ close to J_{H1} there is none;
- (b) (supercritical) for $J < J_{H2}$ close to J_{H2} , where the equilibrium is unstable, there is an orbitally asymptotically stable limit cycle; for $J \geq J_{H2}$ close to J_{H2} there is none.

The corollary is local: it gives no bound on how far from J_{Hi} the cycles persist. That the saddle-type orbit of Theorem 5(b) at $J = 8$ belongs to the family born at J_{H1} is numerical evidence only (Section 8).

Theorem 4 (computer-assisted). *Let $J = 8$. For every $E_l \in [10.59, 10.62]$ the system (1) has a periodic orbit $\Gamma(E_l)$ that crosses Σ_{20} with $\dot{u} > 0$ once per period, whose minimal period lies in $[16.005827509, 16.013912063]$ ms, along which u reaches at least 95.953 mV, and whose nontrivial*

Floquet multipliers satisfy $|\mu| \leq 0.5446$. It is orbitally asymptotically stable. Its point on Σ_{20} is the only fixed point of the first-return map in a box with half-widths at most about $4.6 \cdot 10^{-7}$ that contains it (Section 6).

So the firing frequency $1000/T$ lies in $[62.445, 62.478]$ Hz. Figure 3(a) shows the period enclosures.

Theorem 5 (computer-assisted). *Let $J = 8$ and let E_l be one of 10.613, E_l^* and 10.599. At E_l^* the statement holds for every E_l in the program's ball about E_l^* , of radius below 10^{-25} , which Arb prints as the decimal ball $[10.5989209693916785221988785 \pm 6.21 \cdot 10^{-26}]$ (a larger ball that contains it).*

- (a) *There is a periodic orbit Γ_s through Σ_{20} , crossed with $\dot{u} > 0$, with minimal period T_s and nontrivial multipliers μ_1, μ_2, μ_3 as in Table 1; μ_1 is enclosed in a disc that holds exactly one multiplier. u reaches at least 95.953532, 95.953707 and 95.953706 mV along Γ_s (in the order of the three values of E_l). Γ_s is orbitally asymptotically stable.*
- (b) *There is a periodic orbit Γ_u through Σ_5 , crossed with $\dot{u} > 0$, with minimal period T_u and multipliers as in Table 2: μ_1 is real and larger than 1, $|\mu_2| < 1$ and $|\mu_3| < 1$. Γ_u is of saddle type and not orbitally stable, and $u \leq 11.045813, 11.073821$ and 11.073666 mV along it.*
- (c) *Γ_s, Γ_u and the equilibrium are three distinct invariant sets. The section point of Γ_s is the only fixed point of the first-return map in a box with half-widths at most about $1.21 \cdot 10^{-15}$ that contains it, and that of Γ_u in a box with half-widths from about $2.03 \cdot 10^{-15}$ to $2.13 \cdot 10^{-14}$ (at $E_l = 10.613$; these are the Krawczyk boxes of the proof, and Section 6 gives them at the other two values). Table 3 gives enclosures of the section points, not boxes of uniqueness. Nothing is claimed about other periodic orbits.*

Corollary 6 (computer-assisted). *For $J = 8$ and every $E_l \in [10.59, 10.62]$, equivalently for $E_l = 10.613$ and every $J \in [7.9931, 8.0021]$, the system (1) has a locally asymptotically stable equilibrium and an orbitally asymptotically stable periodic orbit. At $E_l = 10.613, E_l^*$ and 10.599 it also has a periodic orbit of saddle type.*

Corollary 7 (computer-assisted). *Let $J = 8$ and let E_l be 10.613, 10.599 or a value in the ball about E_l^* of Theorem 5. The stable orbit Γ_s of Theorem 5(a) is the orbit $\Gamma(E_l)$ of Theorem 4: its section point lies in the interior of the box of Theorem 4 for each of the pieces of the E_l interval that contain E_l (two pieces for 10.613 and 10.599, which are ends of pieces, and one for E_l^*).*

Remark 3.2 (what is not claimed). We do not claim that the equilibrium and $\Gamma(E_l)$ are the only attractors, nor anything about their basins, nor that Γ_u lies on the boundary between the basins; numerically it does lie on a separatrix [31, p. 62]. The orbits are unique only in their boxes; at the three values of Theorem 5 the stable orbits of Theorems 4 and 5 are the same (Corollary 7), and at other values of E_l we have not compared them with anything. Nothing is proved about periodic orbits at currents other than $J = 8$ (and those equivalent to it by Lemma 2.2), about the fold of cycles that numerically bounds the bistable range from below, or about the equilibria for $J \geq 4089.4815$.

Table 1: Theorem 5(a), the stable orbit Γ_s on Σ_{20} at $J = 8$: the minimal period (ms) and the nontrivial Floquet multipliers. Every bound is rounded outward from the program's balls (`data/certify_bistability.txt`, stages 4 and 7).

E_l	T_s	μ_1	bound on $ \mu_2 , \mu_3 $
10.613	[16.00771280383, 16.00771280384]	$ \mu_1 - 0.070896780693 \leq 2.25 \cdot 10^{-9}$	$3.97 \cdot 10^{-9}$
E_l^*	[16.01150429806, 16.01150429807]	$ \mu_1 - 0.070904264160 \leq 2.26 \cdot 10^{-9}$	$3.99 \cdot 10^{-9}$
10.599	[16.01148299894, 16.01148299895]	$ \mu_1 - 0.070904221789 \leq 2.26 \cdot 10^{-9}$	$3.99 \cdot 10^{-9}$

Table 2: Theorem 5(b), the saddle-type orbit Γ_u on Σ_5 at $J = 8$.

E_l	T_u	μ_1 (real)	μ_2	bound on $ \mu_3 $
10.613	[14.34360094, 14.34360095]	[10.30268246, 10.30308318]	$ \mu_2 - 0.1968322291 \leq 2.72 \cdot 10^{-4}$	$1.57 \cdot 10^{-4}$
E_l^*	[14.36944974, 14.36944975]	[10.52978728, 10.53004150]	$ \mu_2 - 0.1971756106 \leq 1.78 \cdot 10^{-4}$	$9.49 \cdot 10^{-5}$
10.599	[14.36930333, 14.36930334]	[10.52836245, 10.52886553]	$ \mu_2 - 0.1971736153 \leq 3.51 \cdot 10^{-4}$	$1.88 \cdot 10^{-4}$

Table 3: Enclosures of the section points (m, n, h) of Theorem 5: each lies in the product of the three intervals, an outward rounding of the set K of the proof (Section 6). Uniqueness is proved only in the Krawczyk boxes of Theorem 5(c), which are about 10^2 to 10^4 times narrower.

orbit, E_l	m	n	h
Γ_s , 10.613	[0.24052906069, 0.24052906070]	[0.42490470823, 0.42490470824]	[0.39231037641, 0.39231037642]
Γ_s , E_l^*	[0.24053257124, 0.24053257125]	[0.42490307734, 0.42490307735]	[0.39231449643, 0.39231449644]
Γ_s , 10.599	[0.24053255147, 0.24053255148]	[0.42490308639, 0.42490308640]	[0.39231447350, 0.39231447351]
Γ_u , 10.613	[0.09004142752, 0.09004142753]	[0.37950318996, 0.37950318997]	[0.43754826309, 0.43754826310]
Γ_u , E_l^*	[0.09003971888, 0.09003971889]	[0.37946676035, 0.37946676036]	[0.43755923369, 0.43755923370]
Γ_u , 10.599	[0.09003972849, 0.09003972850]	[0.37946696511, 0.37946696512]	[0.43755917467, 0.43755917468]

4 Proofs for the branch of equilibria

4.1 Lemmas

Lemma 4.1 (proved). *Let $p(\lambda) = \lambda^4 + a_1\lambda^3 + a_2\lambda^2 + a_3\lambda + a_4$ be real with $a_1, a_3, a_4 > 0$, $D_2 = a_1a_2 - a_3$ and $D_3 = a_3D_2 - a_1^2a_4$.*

- (a) *p has a root on the imaginary axis if and only if $D_3 = 0$. In that case, with $\omega = \sqrt{a_3/a_1} > 0$, $p(\lambda) = (\lambda^2 + \omega^2)(\lambda^2 + a_1\lambda + a_4/\omega^2)$, the roots $\pm i\omega$ are simple and the other two roots have negative real part.*
- (b) *If $D_3 > 0$, then $D_2 > 0$ and every root has negative real part.*
- (c) *If $D_3 < 0$, then exactly two roots, counted with multiplicity, have positive real part.*

Proof. (a) $p(0) = a_4 \neq 0$. For real $\omega \neq 0$, $p(i\omega) = \omega^4 - a_2\omega^2 + a_4 + i\omega(a_3 - a_1\omega^2)$, so $p(i\omega) = 0$ if and only if $\omega^2 = a_3/a_1$ and $\omega^4 - a_2\omega^2 + a_4 = 0$; with $\omega^2 = a_3/a_1$ the second condition reads $(a_3^2 - a_1a_2a_3 + a_1^2a_4)/a_1^2 = -D_3/a_1^2 = 0$. Expanding $(\lambda^2 + \omega^2)(\lambda^2 + a_1\lambda + c)$ gives $\lambda^4 + a_1\lambda^3 + (c + \omega^2)\lambda^2 + a_1\omega^2\lambda + c\omega^2$; with $c = a_4/\omega^2$ the last two coefficients are a_3 and a_4 , and $c + \omega^2 = a_2$ is equivalent to $a_1^2a_4 + a_3^2 = a_1a_2a_3$, that is $D_3 = 0$. The quadratic $\lambda^2 + a_1\lambda + a_4/\omega^2$ has positive coefficients, so its roots have negative real part; it does not vanish at $\pm i\omega$, so these roots are simple.

(b), (c) If $D_3 > 0$ then $a_3D_2 = D_3 + a_1^2a_4 > 0$, so $D_2 > 0$. Let $Q = \{(a_1, \dots, a_4) : a_1, a_3, a_4 > 0\}$ and $N(a)$ the number of roots of p in $\{\text{Re } \lambda > 0\}$. The roots of a monic polynomial depend continuously on its coefficients, so N is locally constant on the set of coefficient vectors for which

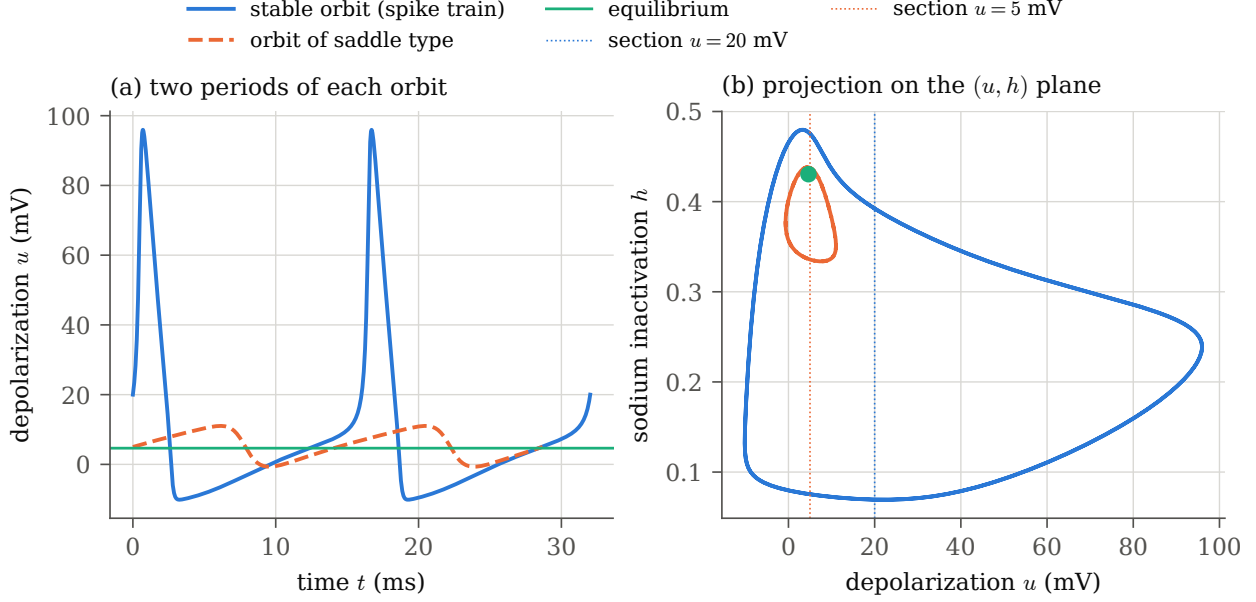


Figure 2: The orbits of Theorem 5 at $J = 8$, $E_l = 10.613$, integrated in floating point from the high-precision section points of Section 8 (code/hh_make_figures.py). (a) $u(t)$ over two periods of the stable orbit Γ_s (solid) and of the saddle-type orbit Γ_u (dashed), and the equilibrium u^* . (b) The projection on the (u, h) plane, with the sections $u = 20$ and $u = 5$ (dotted) and the equilibrium (dot). The enclosures of the theorems are far narrower than the lines.

no root lies on the imaginary axis (a root can enter or leave the open right half-plane only through the axis). By (a) this set, within Q , is $\{D_3 \neq 0\}$. Now $D_3 > 0$ is equivalent to $a_2 > \theta(a_1, a_3, a_4) := a_3/a_1 + a_1 a_4/a_3$, a continuous function on $(0, \infty)^3$, so $Q \cap \{D_3 > 0\}$ is connected, and so is $Q \cap \{D_3 < 0\} = \{a_2 < \theta\}$. Hence N is constant on each. $(\lambda + 1)^4$ has $(a_1, a_2, a_3, a_4) = (4, 6, 4, 1)$, $D_3 = 64 > 0$ and $N = 0$; $(\lambda^2 - \lambda + 1)(\lambda^2 + 3\lambda + 1) = \lambda^4 + 2\lambda^3 - \lambda^2 + 2\lambda + 1$ has $D_3 = -12 < 0$ and $N = 2$, the roots of $\lambda^2 - \lambda + 1$ having real part $1/2$ and those of $\lambda^2 + 3\lambda + 1$ being negative. $\square$

Lemma 4.2 (proved). *If every eigenvalue of $A = Df(x_0)$ at an equilibrium x_0 of a C^2 vector field has negative real part, x_0 is locally asymptotically stable.*

Proof. There are $K, \gamma > 0$ with $\|e^{At}\| \leq Ke^{-\gamma t}$, so $P = \int_0^\infty e^{A^T t} e^{At} dt$ exists, is symmetric positive definite and satisfies $A^T P + PA = -I$ (integrate $\frac{d}{dt} e^{A^T t} e^{At}$). Write $f(x_0 + y) = Ay + g(y)$ with $|g(y)| \leq L|y|^2$ for $|y| \leq 1$ and put $V(y) = y^T P y$. Along solutions $\dot{V} = -|y|^2 + 2y^T P g(y) \leq -|y|^2/2$ for $|y| \leq r_0 = \min(1, 1/(4L\|P\|))$. With $\lambda_{\min}|y|^2 \leq V \leq \lambda_{\max}|y|^2$ (λ the extreme eigenvalues of P), the set $\{V < \lambda_{\min} r_0^2\}$ lies in $\{|y| < r_0\}$, is positively invariant, and on it $\dot{V} \leq -V/(2\lambda_{\max})$, so $V(y(t)) \leq V(y(0))e^{-t/(2\lambda_{\max})} \rightarrow 0$. This gives stability and attraction. $\square$

Lemma 4.3 (proved). *If $A = Df(x_0)$ at an equilibrium x_0 of a C^2 vector field has an eigenvalue with positive real part, x_0 is not stable.*

Proof. Let $4\gamma > 0$ be the smallest positive real part of an eigenvalue of A , and split $\mathbb{R}^n = E_+ \oplus E_-$ into the invariant subspaces of the eigenvalues with positive and with nonpositive real part, $A = \text{diag}(A_+, A_-)$ in adapted coordinates $y = (y_+, y_-)$. For a real matrix all of whose eigenvalues have real part at least 4γ (at most 0) there is an inner product with $\langle A_+ v, v \rangle \geq 3\gamma|v|^2$ (with $\langle A_- v, v \rangle \leq \gamma|v|^2$): use a basis in which the matrix is in real Jordan form with off-diagonal entries scaled below γ . With these inner products on $E_\pm$, put $V(y) = |y_+|^2 - |y_-|^2$ and write $f(x_0 + y) = Ay + g(y)$,

$|g(y)| \leq L|y|^2$. Then $\dot{V} \geq 6\gamma|y_+|^2 - 2\gamma|y_-|^2 - 4L|y|^3$. On the cone $\{V > 0\}$, $|y| \leq \sqrt{2}|y_+|$, so $\dot{V} \geq 4\gamma|y_+|^2 - 4\sqrt{8}L|y_+|^3 \geq 2\gamma|y_+|^2 \geq 2\gamma V$ when $|y| \leq r_0$ for a suitable $r_0 > 0$. A solution starting in the cone with $|y(0)|$ arbitrarily small stays in it while $|y| \leq r_0$, and V grows at least like $e^{2\gamma t}$ there, so it leaves the ball of radius r_0 . $\square$

Lemma 4.4 (proved). (a) $|B_n|/n! \leq 4/(2\pi)^n$ for every $n \geq 0$. Hence, for a real ball X with $|x| \leq r < 3$ on X and $0 \leq k < N + 2 - (N + 2)r/(2\pi)$, the k -th Taylor coefficient of Ψ at $x \in X$ is $\sum_{n=k}^N \frac{B_n}{n!} \binom{n}{k} x^{n-k} + \theta$ with $|\theta| \leq \frac{4 \binom{N+1}{k} r^{N+1-k}}{(2\pi)^{N+1}(1-\rho)}$, $\rho = \frac{N+2}{N+2-k} \cdot \frac{r}{2\pi}$. (b) $E(w) = (e^w - 1)/w = 1/\Psi(w)$ is entire and positive on $\mathbb{R}$, and its Taylor coefficients $e_j(w) = E^{(j)}(w)/j!$ satisfy $e_j(w) = \frac{1}{j!} \int_0^1 s^j e^{sw} ds = {}_1F_1(j+1; j+2; w)/(j+1)!$. Each e_j is strictly increasing in w , so on an interval $[w_-, w_+]$ it lies between $e_j(w_-)$ and $e_j(w_+)$.

Proof. (a) $\Psi(x) = \sum_n B_n x^n/n!$ for $|x| < 2\pi$ [6, eq. 24.2.1] with $B_0 = 1$, $B_1 = -1/2$ and $B_n = 0$ for odd $n \geq 3$ [6, Sect. 24.2]. For even $n \geq 2$, $|B_n|/n! = 2\zeta(n)/(2\pi)^n$ [6, eq. 25.6.2] and $\zeta(n) \leq \zeta(2) = \pi^2/6$, so $|B_n|/n! \leq (\pi^2/3)/(2\pi)^n < 4/(2\pi)^n$; for $n = 0, 1$ the bound reads $1 \leq 4$ and $1/2 \leq 2/\pi$. Differentiating the series k times, the tail $n > N$ of the k -th coefficient is bounded by $\sum_{n>N} 4 \binom{n}{k} r^{n-k}/(2\pi)^n$, whose ratio of consecutive terms, $\frac{n+1}{n+1-k} \cdot \frac{r}{2\pi}$, does not increase with n and is at most $\rho < 1$ for $n \geq N + 1$; a geometric series gives the bound. (b) $E(w) = \int_0^1 e^{sw} ds$ is entire and positive for real w , so $e_j(w) = \frac{1}{j!} \int_0^1 s^j e^{sw} ds$, and its derivative $\frac{1}{j!} \int_0^1 s^{j+1} e^{sw} ds$ is positive. Expanding e^{sw} , $\int_0^1 s^j e^{sw} ds = \sum_{k \geq 0} \frac{w^k}{k!(j+k+1)}$, while ${}_1F_1(j+1; j+2; w) = \sum_{k \geq 0} \frac{(j+1)_k}{(j+2)_k} \frac{w^k}{k!} = \sum_{k \geq 0} \frac{j+1}{j+1+k} \frac{w^k}{k!}$ by the definition of Kummer's function [6, eq. 13.2.2]. $\square$

The program for Theorems 1 and 2 (`code/hh_ball.py`) evaluates Ψ through (a) with $N = 240$ where $|x| < 3$ and through its closed form elsewhere, where it refuses any ball containing 0; the program for Theorems 4 and 5 (`code/hh_arb.py`) evaluates the Taylor coefficients of E through (b), with ${}_1F_1$ from Arb [23], and those of Ψ as the reciprocal series. The two implementations give overlapping enclosures of E_l^* (the self-test of `data/certify_equilibria_hopf.txt`, part 0).

The first Lyapunov coefficient. Let A have a simple pair of eigenvalues $\pm i\omega$, $\omega > 0$, let $Aq = i\omega q$, $A^T p = -i\omega p$ and $\langle p, q \rangle = \bar{p}^T q = 1$, and write the Taylor expansion of the vector field at the equilibrium x_0 as $f(x_0 + y) = Ay + \frac{1}{2}B(y, y) + \frac{1}{6}C(y, y, y) + O(|y|^4)$ with symmetric multilinear B and C . Then

$$\ell_1 = \frac{1}{2\omega} \operatorname{Re} \left(\langle p, C(q, q, \bar{q}) \rangle - 2\langle p, B(q, A^{-1}B(q, \bar{q})) \rangle + \langle p, B(\bar{q}, (2i\omega I - A)^{-1}B(q, q)) \rangle \right), \quad (3)$$

as given by Kuznetsov [26, Sect. “First Lyapunov Coefficient”], who cites his book [25] for it; the bifurcation theorem itself goes back to Hopf [20], cited here as a credit only. We use it through the following theorem.

Theorem H (cited: [26, Sects. “Two-dimensional Case”, “Multi-dimensional Case” and “First Lyapunov Coefficient”]). *Let $\dot{x} = f(x, \alpha)$, $x \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$, with f smooth, have a family of equilibria $x^0(\alpha)$ whose Jacobian $A(\alpha)$ has one pair of complex eigenvalues $\mu(\alpha) \pm i\omega(\alpha)$ with $\mu(0) = 0$, $\omega(0) = \omega_0 > 0$, and at $\alpha = 0$ a simple pair $\pm i\omega_0$, n_s eigenvalues with negative and n_u with positive real part, $n_s + n_u + 2 = n$. If $\ell_1(0) \neq 0$ (computed by (3) at $\alpha = 0$) and $\mu'(0) \neq 0$, the system is locally topologically equivalent near the origin to the suspension of the normal form*

$$\dot{y}_1 = \beta y_1 - y_2 + \sigma y_1(y_1^2 + y_2^2), \quad \dot{y}_2 = y_1 + \beta y_2 + \sigma y_2(y_1^2 + y_2^2), \quad \dot{y}^s = -y^s, \quad \dot{y}^u = y^u,$$

$y^s \in \mathbb{R}^{n_s}$, $y^u \in \mathbb{R}^{n_u}$, $\sigma = \text{sign } \ell_1(0)$. For $\sigma = +1$ the origin of the planar normal form is asymptotically stable for $\beta < 0$ and unstable for $\beta \geq 0$, and a unique limit cycle, unstable, exists for $\beta < 0$; for $\sigma = -1$ the origin is asymptotically stable for $\beta \leq 0$ and unstable for $\beta > 0$, and a unique limit cycle, stable, exists for $\beta > 0$.

The implementation of (3) is tested on systems whose ℓ_1 is known independently: the planar formula of Guckenheimer and Holmes as printed in [26], and the following four-dimensional system, whose ℓ_1 we compute through the centre manifold.

Lemma 4.5 (proved). *For $\omega, a, e > 0$ and real ς, b, c, d, ϕ , the system*

$$\begin{aligned} \dot{x}_1 &= -\omega x_2 + \varsigma x_1 r^2 + c x_1 x_3 + \phi x_1 x_4, & \dot{x}_3 &= -a x_3 + b r^2, \\ \dot{x}_2 &= \omega x_1 + \varsigma x_2 r^2 + c x_2 x_3 - \phi x_2 x_4, & \dot{x}_4 &= -e x_4 + d(x_1^2 - x_2^2), \end{aligned} \quad r^2 = x_1^2 + x_2^2,$$

has, at the origin and with $q = p = (1, -i, 0, 0)/\sqrt{2}$, the first Lyapunov coefficient $\ell_1 = \frac{2}{\omega}(\varsigma + \frac{bc}{a} + \frac{d\phi e}{2(e^2 + 4\omega^2)})$, that is, $\ell_1 = \text{Re } c_1/\omega$ where c_1 is the coefficient of $w|w|^2$ in the equation on the centre manifold in the coordinate w with $x = wq + \bar{w}\bar{q} + O(|w|^2)$.

Proof. With $z = x_1 + ix_2$, the linear part is $\dot{z} = i\omega z$ on the centre subspace. A centre manifold is a graph $x_3 = h_3(z)$, $x_4 = h_4(z)$ with $h_{3,4} = O(|z|^2)$; its quadratic parts solve the homological equations. For $h_3 = \eta_3|z|^2$: $\frac{d}{dt}|z|^2 = 0$ along the linear flow, so $-a\eta_3|z|^2 + b|z|^2 = 0$, $\eta_3 = b/a$. For $h_4 = \text{Re}(\kappa z^2)$: $\frac{d}{dt}\text{Re}(\kappa z^2) = \text{Re}(2i\omega\kappa z^2)$ along the linear flow and $x_1^2 - x_2^2 = \text{Re}(z^2)$, so $2i\omega\kappa = -e\kappa + d$, $\kappa = d/(e + 2i\omega)$. On the manifold, $\dot{z} = \dot{x}_1 + i\dot{x}_2 = i\omega z + \varsigma z|z|^2 + czh_3 + \phi \bar{z}h_4 + O(|z|^4)$, because $x_1x_4 - ix_2x_4 = \bar{z}x_4$, and the terms of $h_{3,4}$ of order three contribute only at order four. Since $\phi \bar{z} \text{Re}(\kappa z^2) = \frac{\phi}{2}(\kappa z^2 \bar{z} + \bar{\kappa} \bar{z}^3)$, the coefficient of $z^2 \bar{z}$ is $\varsigma + cb/a + \phi\kappa/2$. There are no quadratic terms, so this is the resonant cubic coefficient in z . With $x = wq + \bar{w}\bar{q}$ on the centre subspace we have $z = \sqrt{2}w$, so the coefficient in w is $c_1 = 2(\varsigma + cb/a + \phi\kappa/2)$, and $\text{Re } c_1/\omega = \frac{2}{\omega}(\varsigma + \frac{bc}{a} + \frac{\phi}{2} \text{Re } \kappa)$ with $\text{Re } \kappa = de/(e^2 + 4\omega^2)$. Evaluating (3) directly gives the same value: $\langle p, C(q, q, \bar{q}) \rangle = 4\varsigma$, $B(q, \bar{q}) = (0, 0, 2b, 0)$, $\langle p, B(q, A^{-1}B(q, \bar{q})) \rangle = -2bc/a$, $B(q, q) = (0, 0, 0, 2d)$ and $\langle p, B(\bar{q}, (2i\omega - A)^{-1}B(q, q)) \rangle = 2d\phi/(e + 2i\omega)$. $\square$

The program runs the implementation of (3) on two planar test systems and on the system of Lemma 4.5 with $(\omega, \varsigma, a, b, c, d, e, \phi) = (1, -\frac{1}{10}, 2, \frac{1}{3}, \frac{1}{2}, \frac{3}{2}, \frac{1}{2}, 1)$ and with $\omega = \frac{3}{2}$, $\varsigma = -\frac{1}{5}$; the known values of ℓ_1 are $[-0.160416666667, -0.160416666666]$, $[0.125573192239, 0.125573192240]$, $[0.143137254901, 0.143137254902]$ and $[-0.101501501502, -0.101501501501]$, of both signs, and the implementation reproduces each to within 10^{-60} . Two mutated versions, with $+2$ in place of -2 in the middle term and with $(-A)^{-1}$ in place of $(2i\omega I - A)^{-1}$, give $[-0.19019608, -0.19019607]$ and $[2.96666666, 2.96666667]$ on the first coupled system and are refused (negative controls).

4.2 Proofs

All references to checks are to `data/certify_equilibria_hopf.txt`, written by `code/certify_equilibria_hopf.py` at 256 bits; the model is `code/hh_ball.py`, which evaluates every quantity as a truncated power series in u over complex balls, so that a series evaluated on a ball U encloses the Taylor coefficients of the quantity at every point of U . The leak potential enters as the ball $[10.605 \pm 0.015] \supset [10.59, 10.62]$.

Proof of Proposition 2.3. $I(x^*(0)) = 0$ is linear in E_l and gives the formula. The program evaluates it in ball arithmetic and prints the resulting ball, of radius below 10^{-38} (part 0 of the output, the line that begins “E_l* =”, which also gives the current at $u = 0$); that ball is the step of the proof, and

[10.5989209693916785221988785, 10.5989209693916785221988786] is its outward rounding. The ball computed by the independent implementation of `code/hh_arb.py` overlaps it; that comparison is a self-test, not a step of the proof. The current at $u = 0$ for $E_l = 10.613$ is $I(x^*(0)) = g_l(E_l^* - 10.613)$, since I depends on E_l only through $-g_l E_l$. $\square$

Proof of Theorem 1. By Lemma 2.1(d) we must count the solutions of $J_{ss}(u) = J$. For $u < -12$ each of the three terms of $J_{ss}(u) = g_{Na}m^3h(u - 115) + g_Kn^4(u + 12) + g_l(u - E_l)$ is negative (Lemma 2.1(a) and $E_l > -12$), so $J_{ss}(u) < 0 \leq J$. For $u \geq 115$ the first term is nonnegative, the third positive, and the second at least $g_Kn_\infty(115)^4 \cdot 127$ by Lemma 2.1(b); the program encloses this number and finds it at least $4089.4815 > 200$ (part A, third check). On $[-12, 115]$ the program shows $J'_{ss} > 0$ on each of 74 pieces of a subdivision (part A, first check), so J_{ss} is strictly increasing there, and it shows $J_{ss}(-12) \leq -6.7999 < 0$ and $J_{ss}(115) > 200$ (fourth check) for every E_l in the ball. For $J \in [0, 200]$ the intermediate value theorem and monotonicity give exactly one solution, and it lies in $(-12, 115)$. J_{ss} is real-analytic with positive derivative there, so its inverse u^* is real-analytic and increasing. $\square$

Proof of Theorem 2. Signs along the branch. The program subdivides $[-12, 115]$ into 1043 pieces and shows on each that $a_1, a_3, a_4 > 0$ and that $D_3 > 0$ or $D_3 < 0$, except on 52 pieces of width below 10^{-5} (part B, first check). These undecided pieces form two clusters, $[5.345749617, 5.345961571]$ and $[21.941817522, 21.941999197]$ (second check); every zero of D_3 in $[-12, 115]$ lies in them. On each cluster, slightly enlarged, the program encloses dD_3/du in $[-4.121, -4.109]$ and $[176.1, 176.6]$ respectively and shows that D_3 has opposite signs at the two ends (third and fifth checks), so each cluster contains exactly one zero, and it is simple. Interval Newton steps $N(X) = m - D_3(m)/D'_3(X)$, m the midpoint of X , are applied as long as $N(X) \subset X$; by the mean value theorem the zero $u_H \in X$ satisfies $u_H = m - D_3(m)/D'_3(\xi)$ for some $\xi \in X$, so it stays in the new box, which gives u_{H1} and u_{H2} in balls of radius at most $2.28 \cdot 10^{-73}$ and $3.69 \cdot 10^{-73}$ (fourth and sixth checks; Theorem 2 prints them rounded outward to 12 decimals). The program shows $D_3 > 0$ at $u = -12$ and $u = 115$ and $D_3 < 0$ at a point between the zeros (seventh check), which fixes the signs on the three intervals.

(a) and (b). By Theorem 1 the equilibrium at current J has $u^*(J) \in (-12, 115)$, and $u^*(J) < u_{H1}$ exactly when $J < J_{H1}$, since u^* is increasing; similarly for u_{H2} . Lemma 4.1(b) gives the stable ranges, and Lemma 4.2 local asymptotic stability; Lemma 4.1(c) and (a) give the count on (J_{H1}, J_{H2}) and the absence of roots on the axis, and Lemma 4.3 the instability; Lemma 4.1(a) gives (b), with $\omega_i^2 = a_3/a_1$ at u_{Hi} , enclosed by the program. The program also checks, as a guard against coding errors, that $a_2 - \omega^2 - a_4/\omega^2$ encloses 0 at u_{Hi} and that the Routh array at the midpoint of (u_{H1}, u_{H2}) has two sign changes; both are consequences of Lemma 4.1, not steps of the proof.

(c). $J_{Hi} = J_{ss}(u_{Hi})$ is evaluated on the ball of u_{Hi} for each value of E_l , and for E_l^* on its ball; Lemma 2.2 gives the affine dependence and the range over $[10.59, 10.62]$, whose ends are the values at 10.62 and 10.59 (part C).

(d). The coefficients a_k are real-analytic in u , and $i\omega_i$ is a simple root of $p(\cdot; u_{Hi})$, so by the implicit function theorem there is an analytic root $\lambda(u)$ with $\lambda(u_{Hi}) = i\omega_i$ and $\lambda'(u) = -\partial_u p / \partial_{\lambda} p$. The program encloses $\text{Re } \lambda'(u_{Hi})$ in $[0.0501908689, 0.0501908690]$ and $[-0.0824086729, -0.0824086728]$, and $J'_{ss}(u_{Hi})$ in $[2.67128168, 2.67128169]$ and $[18.34451285, 18.34451286]$; the eigenvalue as a function of J is $\lambda(u^*(J))$, with derivative $\lambda'(u)/J'_{ss}(u)$, enclosed as stated.

(e). By Lemma 2.4(c) with $\lambda = i\omega_i$, the closed-form vectors q and w satisfy $Aq = i\omega q$ and $w^T A = i\omega w^T$ (the program also checks that the residuals enclose 0). The program normalizes $\langle q, q \rangle = 1$ and $w^T q = 1$, and puts $p = \bar{w}$, so that $A^T p = -i\omega p$ and $\langle p, q \rangle = 1$. It obtains B and C from the vector field evaluated as a power series in t along $x^*(u_{Hi}) + tv$: the coefficients of t^2 and t^3

are $\frac{1}{2}B(v, v)$ and $\frac{1}{6}C(v, v, v)$, and $B(v, v')$, $C(q, q, \bar{q})$ follow by polarization. A^{-1} and $(2i\omega I - A)^{-1}$ are applied by Arb's rigorous linear solver. ℓ_1 does not depend on J or E_l , which enter f additively. The resulting enclosures exclude 0 (part C). A non-rigorous recomputation with SymPy derivatives and mpmath eigenvectors agrees to a relative $3.23 \cdot 10^{-40}$ and $6.24 \cdot 10^{-41}$. $\square$

Proof of Corollary 3. Fix E_l and i , and put $\alpha = J - J_{H_i}$, $x^0(\alpha) = x^*(u^*(J))$, which is an analytic family of equilibria by Theorem 1. By Theorem 2(b) the Jacobian at $\alpha = 0$ has a simple pair $\pm i\omega_i$, $n_s = 2$ and $n_u = 0$; the pair continues analytically as $\mu(\alpha) \pm i\omega(\alpha)$ with $\mu'(0) = \operatorname{Re} \lambda'(J_{H_i}) \neq 0$ by (d), and $\ell_1(0) \neq 0$ by (e). Theorem H applies with $\sigma = +1$ at $H1$ and $\sigma = -1$ at $H2$. The topological equivalence of the families maps equilibria to equilibria, preserves their asymptotic stability, maps periodic orbits to periodic orbits, and preserves orbital asymptotic stability and its failure, since these are defined through neighbourhoods and limits along orbits. At $H1$ the equilibrium is asymptotically stable exactly for $J < J_{H1}$ near J_{H1} (Theorem 2(a)), and in the normal form with $\sigma = +1$ exactly for $\beta < 0$; so these parameter sets correspond, and the unstable cycle of the normal form for $\beta < 0$ gives (a); the suspension by $\dot{y}^s = -y^s$ keeps it not orbitally stable. At $H2$ the equilibrium is unstable exactly for $J < J_{H2}$ near J_{H2} and, with $\sigma = -1$, the origin exactly for $\beta > 0$; the stable cycle there, attracting also in the y^s directions, gives (b). $\square$

5 Validated integration and Poincaré maps

This section states and proves what the program `code/certify_bistability.py` relies on; the implementation is `code/hh_lohner.py` (integrator and Poincaré maps), `code/certlib.py` (Krawczyk, Gershgorin, the equilibrium) and `code/ball_stable.py` (the contraction on the E_l interval), with the model in `code/hh_arb.py`. The program extends (1) by E_l as a fifth variable with $\dot{E}_l = 0$, so that an interval of E_l is a direction of the initial set. We write F for this analytic vector field on $\mathbb{R}^5$, φ_t for its flow, and, for $y \in \mathbb{R}^5$,

$$x_k(y) = \frac{1}{k!} \frac{d^k}{dt^k} \varphi_t(y) \Big|_{t=0}, \quad A_k(y) = \frac{1}{k!} \frac{d^k}{dt^k} D\varphi_t(y) \Big|_{t=0} = Dx_k(y).$$

The x_k follow from the recursion $x_{k+1} = [F(x(\cdot))]_k / (k+1)$, $[\cdot]_k$ the k -th Taylor coefficient in t , and the A_k from $A_0 = I$, $A_{k+1} = \frac{1}{k+1} \sum_{i \leq k} [DF(x(\cdot))]_i A_{k-i}$ (the variational equation $\dot{V} = DF(x(t))V$). The program evaluates both recursions in ball arithmetic on a box X , with the Taylor coefficients of the rate functions from exact formulas for the exponentials, from Lemma 4.4(b) for Ψ , and from the reciprocal series of $1 + e^{(30-u_0)/10} e^{-(u-u_0)/10}$ for β_h , where u_0 is the voltage about which the series is expanded. By the inclusion property of ball arithmetic this gives boxes $x_k(X) \ni x_k(y)$ and $A_k(X) \ni A_k(y)$ for all $y \in X$. For intervals we write $[0, h]^k = [0, h^k]$.

5.1 One step

Lemma 5.1 (proved; the a priori enclosure). *Let X and Y be boxes, $h > 0$, $p \geq 1$, and suppose $\sum_{k \leq p} x_k(X)[0, h]^k + x_{p+1}(Y)[0, h]^{p+1} \subset \operatorname{int} Y$. Then for every $y \in X$ the solution $\varphi_s(y)$ exists and lies in Y for $s \in [0, h]$, and $\varphi_s(y) \in \sum_{k \leq p} x_k(y)s^k + x_{p+1}(Y)s^{p+1}$.*

Proof. The left side contains X (the term $k = 0$ is X and the others contain 0), so $X \subset \operatorname{int} Y$. Let $y \in X$ and $s_1 = \sup\{s \in [0, h] : \varphi_\sigma(y) \in Y \text{ for } \sigma \in [0, s]\}$. For $s \leq s_1$ Taylor's theorem with the Lagrange remainder, applied to each component of the analytic function $t \mapsto \varphi_t(y)$, gives $\varphi_s(y)_i = \sum_{k \leq p} x_k(y)_i s^k + x_{p+1}(\varphi_\xi(y))_i s^{p+1}$ for some $\xi \in (0, s)$, because $\frac{1}{(p+1)!} \frac{d^{p+1}}{dt^{p+1}} \varphi_t(y) \Big|_{t=\xi} = x_{p+1}(\varphi_\xi(y))$ by the group property. Since $\varphi_\xi(y) \in Y$, $\varphi_s(y)$ lies in the left side of the hypothesis, hence in $\operatorname{int} Y$.

If $s_1 < h$, the solution would stay in Y a little beyond s_1 (it is in the open set $\text{int } Y$ at s_1), a contradiction; so $s_1 = h$ and the solution, which stays in a compact set, exists on $[0, h]$. $\square$

Lemma 5.2 (proved; the variational enclosure). *With X, Y, h, p as in Lemma 5.1, let W be a box of 5×5 matrices with $\sum_{k \leq p} A_k(X)[0, h]^k + A_{p+1}(Y)W[0, h]^{p+1} \subset \text{int } W$ entrywise. Then $D\varphi_s(y) \in W$ and $D\varphi_s(y) \in N(s) := \sum_{k \leq p} A_k(X)s^k + A_{p+1}(Y)W s^{p+1}$ for $y \in X, s \in [0, h]$.*

Proof. Differentiating $D\varphi_{s+\sigma}(y) = D\varphi_\sigma(\varphi_s(y))D\varphi_s(y)$ k times in σ at $\sigma = 0$ gives

$$\frac{1}{k!} \frac{d^k}{ds^k} D\varphi_s(y) = A_k(\varphi_s(y)) D\varphi_s(y).$$

The argument of Lemma 5.1, entrywise, with $\varphi_\xi(y) \in Y$ from that lemma and $D\varphi_\xi(y) \in W$ up to s_1 , proves the claim; $D\varphi_0 = I$ lies in the left side, hence in $\text{int } W$. $\square$

Lemma 5.3 (proved; the mean-value form). *With X convex, $\bar{y} \in X$ and the hypotheses of Lemma 5.1, for every $y \in X$ and $s \in [0, h]$ there is a matrix $\tilde{M}$ in $M(s) := \sum_{k \leq p} A_k(X)s^k$ such that $\varphi_s(y) - \sum_{k \leq p} x_k(\bar{y})s^k - \tilde{M}(y - \bar{y}) \in x_{p+1}(Y)s^{p+1}$.*

Proof. $T(y) = \sum_{k \leq p} x_k(y)s^k$ has $DT(y) = \sum_{k \leq p} A_k(y)s^k$. By the mean value theorem on the segment from $\bar{y}$ to y , $T_i(y) - T_i(\bar{y}) = DT_i(\zeta_i)(y - \bar{y})$ with $\zeta_i \in X$, and the row $DT_i(\zeta_i)$ lies in row i of $M(s)$. Lemma 5.1 bounds $\varphi_s(y) - T(y)$. $\square$

Sets. The program represents a set of initial points as $S = \{\bar{x} + C\rho_0 + B\rho : \rho_0 \in r_0, \rho \in r\}$ with an exact point $\bar{x}$, exact matrices C, B and boxes r_0, r (the doubleton sets of Zgliczyński [39], after Lohner [30]), and a set of matrices as $\{\bar{V} + B_V R : R \in R_V\}$. One step to time s , with $X \supset S$ and $\tilde{M}$ as in Lemma 5.3, computes: $a = \sum_{k \leq p} x_k(\bar{x})s^k + x_{p+1}(Y)s^{p+1}$, $\bar{M}$ = the midpoint of $M(s)$, $C' =$ the midpoint of $\bar{M}C$, $Z = a + (M(s) - \bar{M})(Cr_0 + Br) + (\bar{M}C - C')r_0$, $\bar{x}' =$ the midpoint of Z , a matrix B' from a floating-point QR decomposition, a ball enclosure of B'^{-1} , and $r' = (B'^{-1}\bar{M}B)r + B'^{-1}(Z - \bar{x}')$; the box r_0 is kept. The program keeps $0 \in r_0$ and $0 \in r$: an initial set has $r = 0$ and r_0 a box centred at 0 (the half-widths of the initial box on the section, and the E_l ball minus its midpoint), and the update gives $0 \in r'$ because $0 \in r$ and $0 \in Z - \bar{x}'$, $\bar{x}'$ being the midpoint of Z . Hence $\bar{x} \in S \subset X$, and Lemma 5.3 applies with $\bar{y} = \bar{x}$.

Lemma 5.4 (proved). *If $y = \bar{x} + C\rho_0 + B\rho \in S$, then $\varphi_s(y) = \bar{x}' + C'\rho_0 + B'\rho'$ for some $\rho' \in r'$. The same holds for the matrix sets: if $D\varphi_t(x_0) = \bar{V} + B_V R, R \in R_V$, and $D\varphi_s(\varphi_t(x_0)) \in N(s)$, the program's update gives $D\varphi_{t+s}(x_0) = \bar{V}' + B'_V R'$ with $R' \in R'_V$.*

Proof. By Lemma 5.3, $\varphi_s(y) = \alpha + \tilde{M}(C\rho_0 + B\rho)$ with $\alpha \in a$ and $\tilde{M} \in M(s)$. Then $\varphi_s(y) = \zeta + C'\rho_0 + \bar{M}B\rho$ with $\zeta = \alpha + (\tilde{M} - \bar{M})(C\rho_0 + B\rho) + (\bar{M}C - C')\rho_0 \in Z$, and $\varphi_s(y) = \bar{x}' + C'\rho_0 + B'\rho'$ with $\rho' = B'^{-1}\bar{M}B\rho + B'^{-1}(\zeta - \bar{x}') \in r'$, all inclusions being those of ball arithmetic. For matrices, $\tilde{N}(\bar{V} + B_V R) = \tilde{N}\bar{V} + (\tilde{N} - \bar{N})(\bar{V} + B_V R) + \tilde{N}B_V R$ with $\tilde{N} \in N(s)$ and $\bar{N}$ its midpoint, and the same rearrangement applies. $\square$

5.2 Crossing a section

Let $\Sigma = \{x_i = c\}$ and $g(x) = x_i - c$ (the sections here are crossed with $\dot{x}_i > 0$). The Poincaré routine (`code/hh_lohner.py, poincare`) starts from a set S_0 whose hull lies exactly on Σ (otherwise it stops: `check_on_section`), and integrates step by step, classifying every step by the a priori box Y of the step, an enclosure $F_i(Y)$ of $\dot{x}_i$ on Y , and the hull X of the set at the start of the step:

- (D) *departure*: while the hull meets Σ , every step must have $F_i > 0$ on Y ;
- (N) *no crossing*: $g \neq 0$ on Y ; or $F_i > 0$ on Y and $g > 0$ on X ; or $F_i > 0$ on Y and $g < 0$ on the hulls at both ends of the step;
- (P) *pass*: $F_i < 0$ on Y (counted as a downward pass);
- (H) *hit*: $F_i > 0$ on Y , $g < 0$ on X , and times $0 \leq s_- < s_+ \leq h$ with $g < 0$ on the enclosure of the set at time s_- and $g > 0$ at time s_+ .

Lemma 5.5 (proved; first hit and the crossing). *Suppose a run from $S_0 \subset \Sigma$ consists of steps of types (D), (N), (P) followed by one step of type (H) that ends at the total time $t_0 + s_+$. For $y \in S_0$ let $\tau(y)$ be the smallest $t > 0$ with $g(\varphi_t(y)) = 0$ and $\frac{d}{dt}g(\varphi_t(y)) > 0$. Then $\tau(y)$ exists and lies in $t_0 + (s_-, s_+)$, every $t \in (0, \tau(y))$ with $g(\varphi_t(y)) = 0$ has $\frac{d}{dt}g(\varphi_t(y)) < 0$, and there are at most as many such t as steps of type (P). Moreover, with $Y_T \supset \varphi_{[t_0+s_-, t_0+s_+]}(S_0)$ and any $s_m \in [s_-, s_+]$, the point $P(y) = \varphi_{\tau(y)}(y)$ satisfies, for each j ,*

$$P_j(y) = \varphi_{t_0+s_m}(y)_j - q_j (\varphi_{t_0+s_m}(y)_i - c), \quad \tau(y) = t_0 + s_m - \frac{\varphi_{t_0+s_m}(y)_i - c}{\bar{F}_i},$$

with q_j in the interval of values of F_j/F_i on Y_T and $\bar{F}_i$ in the interval of values of F_i on Y_T .

Proof. On a step with $F_i > 0$ on Y the function $t \mapsto g(\varphi_t(y))$ is strictly increasing, and on a step with $F_i < 0$ strictly decreasing, since $\frac{d}{dt}g(\varphi_t(y)) = F_i(\varphi_t(y))$ and $\varphi_t(y) \in Y$ (Lemma 5.1). In the departure phase $g(\varphi_0(y)) = 0$ and g increases, so $g > 0$ for $t > 0$ until the hull leaves Σ . A step of type (N) has no zero of g : either $g \neq 0$ on Y , or g increases from a positive value, or g increases and is negative at both ends. A step of type (P) has at most one zero, with negative derivative. On the step of type (H), g is negative at the start and increasing, negative at s_- and positive at s_+ , so it has exactly one zero, in (s_-, s_+) , with positive derivative. This proves the first part. For the formulas, $\varphi_{\tau(y)}(y)_j - \varphi_{t_0+s_m}(y)_j = \int F_j dt$ and $c - \varphi_{t_0+s_m}(y)_i = \int F_i dt$ over the arc between the two times, which lies in Y_T ; since $F_i > 0$ on Y_T , the first integral is $\int (F_j/F_i) F_i dt$, a weighted mean of F_j/F_i times the second, and the second is the length of the time interval times a mean of F_i . $\square$

The program evaluates these formulas on the Lohner representation of $\varphi_{t_0+s_m}(S_0)$ (Lemma 5.4), so that the time shift and the displacement along the flow cancel to first order, intersects the result with Y_T , and records the extremes of u over the step enclosures; the “ u reaches at least” bounds of the theorems are maxima over step times of lower bounds of u on the set, so every orbit from S_0 attains them.

Lemma 5.6 (proved; the derivative). *Under the hypotheses of Lemma 5.5, every $y \in S_0$ has a neighbourhood W in Σ and a real-analytic $\tilde{\tau}$ on W with $\tilde{\tau} = \tau$ on $S_0 \cap W$; so τ and P are analytic on S_0 in this sense, and $DP(y) = (I - F(P(y))e_i^T/F_i(P(y)))D\varphi_{\tau(y)}(y)$ on the tangent space of Σ . If $D\varphi_{t_0}(y) \in \bar{V} + B_V R_V$ and $N(T)$ is the enclosure of Lemma 5.2 over $T = [s_-, s_+]$, then $DP(y)$ lies in $(I - F(\mathcal{P})e_i^T/F_i(\mathcal{P}))N(T)(\bar{V} + B_V R_V)$, where $\mathcal{P}$ is the enclosure of $P(S_0)$.*

Proof. $G(t, z) = g(\varphi_t(z))$ has $\partial_t G(\tau(y), y) = F_i(P(y)) > 0$, so the implicit function theorem gives an analytic $\tilde{\tau}$ near y with $\tilde{\tau}(y) = \tau(y)$ and $G(\tilde{\tau}(z), z) = 0$. For $z \in S_0$ near y , $\tilde{\tau}(z)$ lies in the window $t_0 + (s_-, s_+)$ of the step of type (H), where $t \mapsto G(t, z)$ is strictly increasing (proof of Lemma 5.5); its only zero there is $\tau(z)$, so $\tilde{\tau}(z) = \tau(z)$. Differentiating, $D\tilde{\tau} = -e_i^T D\varphi_{\tilde{\tau}}/F_i(P)$ and $DP = D\varphi_{\tilde{\tau}} + F(P)D\tilde{\tau}$. The enclosure follows from $D\varphi_{\tau(y)} = D\varphi_{\tau-t_0}(\varphi_{t_0}(y))D\varphi_{t_0}(y)$ and Lemmas 5.2 and 5.4. $\square$

Since E_l is constant along the flow, Lemmas 5.5 to 5.7 hold as well for the flow of (1) at each fixed E_l of the E_l range of S_0 , with S_0 replaced by its slice at that E_l ; this is how they are applied below. In the coordinates (m, n, h) of Σ_{20} or Σ_5 and for fixed E_l , the matrix of DP is the block of rows and columns m, n, h of the 5×5 matrix of the lemma; its row u vanishes.

Lemma 5.7 (proved; minimal period). *Under the hypotheses of Lemma 5.5, if $y \in S_0$ and $P(y) = y$, the orbit of y is periodic with minimal period $\tau(y)$. If moreover S_0 contains a neighbourhood of y in Σ , then on a neighbourhood of y the map P is the local return map at y of Section 2.*

Proof. $\varphi_{\tau(y)}(y) = y$, and y is not an equilibrium because $F_i(y) > 0$ (departure condition). The periods of a nonconstant periodic orbit are the integer multiples of its minimal period T_0 . If $T_0 < \tau(y)$, then $\varphi_{T_0}(y) = y \in \Sigma$ with $\frac{d}{dt}g = F_i(y) > 0$ at $t = T_0 \in (0, \tau(y))$, which Lemma 5.5 excludes. For the second statement, the function $\tilde{\tau}$ of Lemma 5.6 at y and the return time τ_{loc} of the local return map are both the solution given by the implicit function theorem at $(t, z) = (\tau(y), y)$, which is unique near that point, so they agree near y , and $\tilde{\tau} = \tau$ on S_0 . $\square$

Lemma 5.8 (proved; Floquet multipliers). *Let Γ be a periodic orbit with minimal period T of an analytic vector field F on $\mathbb{R}^d$, through $y \in \Sigma$ with $F_i(y) \neq 0$, let $M = D\varphi_T(y)$, and let P be the local return map at y (Section 2). Then $DP(y) = \Pi M$ on the tangent space $E = \{v : v_i = 0\}$ of Σ , with $\Pi = I - F(y)e_i^T/F_i(y)$, and $\det(\lambda I - M) = (\lambda - 1)\det(\lambda I - DP(y))$.*

Proof. $MF(y) = F(y)$, and $F(y) \notin E$. Differentiating $x_i(\varphi_{\tau_{\text{loc}}(z)}(z)) = c$ at $z = y$, where $\tau_{\text{loc}}(y) = T$ and $\varphi_T(y) = y$, gives $D\tau_{\text{loc}}(y) = -e_i^T M/F_i(y)$ on E , so $DP(y) = M + F(y)D\tau_{\text{loc}}(y) = \Pi M$ on E , where Π is the projection onto E along $F(y)$. For $v \in E$, $Mv = \Pi Mv + \frac{e_i^T Mv}{F_i(y)}F(y)$, so in a basis consisting of $F(y)$ and a basis of E the matrix of M is block upper triangular with diagonal blocks 1 and $DP(y)$. $\square$

5.3 Fixed points, multipliers and stability

Theorem K (cited: [34, Theorem 13.3], a form of the test of Krawczyk [24] and Moore [32]). *Let $f : D \rightarrow \mathbb{R}^n$ be continuously differentiable, $\tilde{x} \in \mathbb{R}^n$, X a box, and R a real $n \times n$ matrix, with $0 \in X$ and $\tilde{x} + X \subset D$. Suppose*

$$S(X, \tilde{x}) := -Rf(\tilde{x}) + (I - RJf(\tilde{x} + X))X \subset \text{int } X,$$

where $Jf(Y)$ is the interval hull of the Jacobians of f at the points of Y . Then R and all matrices $M \in Jf(\tilde{x} + X)$ are nonsingular, and there is a unique root $\hat{x}$ of f in $\tilde{x} + S(X, \tilde{x})$.

This is the theorem as Rump states it, with his notation $Jf(Y) = \text{hull}\{Jf(y) : y \in Y\}$ [34, eq. (13.3)]; the proof there ends by noting that the nonsingularity makes f injective on $\tilde{x} + X$. We use it through the following lemma.

Lemma 5.9 (proved). *Let G be continuously differentiable on a box $\bar{z} + X$ with $0 \in X$, R a real matrix, $\mathcal{G}$ a box containing $G(\bar{z})$ and $\mathbf{M}$ an interval matrix containing $DG(z)$ for every $z \in \bar{z} + X$. If $K := \bar{z} - R\mathcal{G} + (I - R\mathbf{M})X \subset \text{int}(\bar{z} + X)$, then G has exactly one zero in $\bar{z} + X$, and it lies in K .*

Proof. $Jf(\bar{z} + X)$, for $f = G$, is the smallest box of matrices containing every $DG(z)$, so it lies in $\mathbf{M}$, and by the inclusion property of interval arithmetic $\bar{z} + S(X, \bar{z}) \subset K \subset \text{int}(\bar{z} + X)$, that is $S(X, \bar{z}) \subset \text{int } X$. Theorem K with $D = \bar{z} + X$ gives a zero $\hat{x} \in \bar{z} + S(X, \bar{z}) \subset K$ and the nonsingularity of every matrix in $Jf(\bar{z} + X)$. If $G(z_1) = G(z_2)$ with $z_1, z_2 \in \bar{z} + X$, then $0 = \tilde{M}(z_1 - z_2)$ with $\tilde{M} = \int_0^1 DG(z_2 + t(z_1 - z_2)) dt$; each entry of $\tilde{M}$ is a mean of the same entry of Jacobians at points of the convex box $\bar{z} + X$, so $\tilde{M} \in Jf(\bar{z} + X)$ is nonsingular and $z_1 = z_2$. $\square$

The program (`code/certlib.py`, `krawczyk`) applies the lemma to $G = P - \text{id}$ on a box $Z = \bar{z} + X$ of Σ , with R the floating-point inverse of the midpoint of $\mathbf{DP}(Z) - I$: it computes $K = \bar{z} - R(\mathcal{P}(\bar{z}) - \bar{z}) + (I - R(\mathbf{DP}(Z) - I))X$ with an enclosure $\mathcal{P}(\bar{z})$ of $P(\bar{z})$ and an enclosure $\mathbf{DP}(Z)$ of DP over Z , and checks $K \subset \text{int } Z$.

Lemma 5.10 (proved; Gershgorin discs). *Let $\mathbf{M}$ be a real interval matrix, S an invertible complex matrix, and $D_i = \{z : |z - c_i| \leq R_i\}$ discs with $R_i \geq |(S^{-1}MS)_{ii} - c_i| + \sum_{j \neq i} |(S^{-1}MS)_{ij}|$ for every $M \in \mathbf{M}$. Then for every $M \in \mathbf{M}$: (a) every eigenvalue lies in $\bigcup_i D_i$; (b) a union of k discs that is disjoint from the other discs contains exactly k eigenvalues, counted with multiplicity; (c) if D_1 is disjoint from the others and c_1 is real, the eigenvalue in D_1 is real.*

Proof. (a) Gershgorin's theorem [10] for $S^{-1}MS$, whose discs lie in the D_i : if $S^{-1}MSv = \lambda v$ and $|v_i|$ is maximal, $|\lambda - (S^{-1}MS)_{ii}| \leq \sum_{j \neq i} |(S^{-1}MS)_{ij}|$. (b) Let Δ be the diagonal of $S^{-1}MS$ and $M_t = \Delta + t(S^{-1}MS - \Delta)$, $t \in [0, 1]$. The Gershgorin discs of M_t have the same centres and t times the radii, so they lie in the D_i , and by (a) no eigenvalue of M_t crosses the boundary of the union of the k discs. The eigenvalues depend continuously on t ; at $t = 0$ they are the centres of these discs, the diagonal entries of $S^{-1}MS$, and exactly k of them lie in the union. (c) M is real, so $\bar{\lambda}$ is an eigenvalue with $\bar{\lambda} \in D_1$ when $\lambda \in D_1$; by (b) D_1 contains only one eigenvalue, so $\lambda = \bar{\lambda}$. $\square$

The program takes S from the floating-point eigenvectors of the midpoint of $\mathbf{M}$, encloses S^{-1} and $S^{-1}\mathbf{M}S$ in ball arithmetic, and for a real eigenvalue of the midpoint takes a real eigenvector, so that c_1 is real (`code/certlib.py`, `gershgorin_eigenbasis` and `multiplier_test`).

Lemma 5.11 (proved; contraction). *Let $Z \subset \mathbb{R}^3$ be a box, $P : Z \rightarrow \text{int } Z$ of class C^1 with $\|DP(z)\|_\infty \leq \kappa < 1$ for $z \in Z$. Then P has exactly one fixed point z^* in Z , $P^k(z) \rightarrow z^*$ for $z \in Z$, and every eigenvalue of $DP(z^*)$ has modulus at most κ .*

Proof. Z is convex, so $\|P(z) - P(z')\|_\infty \leq \kappa\|z - z'\|_\infty$ by the mean value inequality, and Banach's fixed point theorem applies on the complete set Z . The spectral radius is at most any induced norm. $\square$

Lemma 5.12 (proved; orbital asymptotic stability). *Let Γ be a periodic orbit with minimal period T of an analytic vector field F on $\mathbb{R}^d$, through $y^* \in \Sigma$ with $F_i(y^*) > 0$, let P be the local return map at y^* , and suppose every eigenvalue of $DP(y^*)$ has modulus less than 1. Then Γ is orbitally asymptotically stable.*

Proof. Step 1. There is a norm $|\cdot|_*$ on the tangent space of Σ with $|DP(y^*)|_* < 1$ (for a matrix of spectral radius ρ and any $\varepsilon > 0$ there is an induced norm with value less than $\rho + \varepsilon$). By continuity of DP and the mean value inequality there are $\kappa < 1$ and $\delta_0 > 0$ such that $B_0 = \{z \in \Sigma : |z - y^*|_* \leq \delta_0\}$ is mapped into itself and $|P(z) - y^*|_* \leq \kappa|z - y^*|_*$ there. Shrinking δ_0 , the return time τ_{loc} , which is continuous with $\tau_{\text{loc}}(y^*) = T$, has values in $[T/2, 2T]$ on B_0 .

Step 2. (In the application $d = 4$, at fixed E_l .) By the implicit function theorem applied to $g(\varphi_t(x)) = 0$ at $(t, x) = (0, y^*)$, there are a neighbourhood U of y^* and a continuous t_U on U with $t_U(y^*) = 0$, $|t_U| \leq T/4$ and $\pi(x) = \varphi_{t_U(x)}(x) \in \Sigma$; π is continuous and $\pi(y^*) = y^*$. Given $\eta > 0$ and $\varepsilon > 0$, let $U_\eta \subset U$ be a neighbourhood of y^* with $\pi(U_\eta) \subset B_\eta = \{z \in \Sigma : |z - y^*|_* \leq \eta\}$. The map $(t, x) \mapsto \varphi_t(x)$ is uniformly continuous on $[0, 3T] \times \{x : \text{dist}(x, \Gamma) \leq \delta_1\}$ for a small $\delta_1 > 0$, so there is $\delta \in (0, \delta_1]$ such that $|x - \gamma| < \delta$, $\gamma \in \Gamma$, implies $|\varphi_t(x) - \varphi_t(\gamma)| < \varepsilon$ for $t \in [0, 3T]$ and $\varphi_{2T-s}(x) \in U_\eta$, where $\gamma = \varphi_s(y^*)$, $s \in [0, T]$ (because $\varphi_{2T-s}(\gamma) = y^*$). Such an x stays within ε of Γ up to the time $2T - s + t_U(\varphi_{2T-s}(x)) \in [3T/4, 9T/4]$, at which it lies in B_η .

Step 3. Continuous dependence on initial data on $[0, 2T]$ gives L with $|\varphi_t(z) - \varphi_t(y^*)| \leq L|z - y^*|_*$ for $z \in B_0$, $t \in [0, 2T]$. For $z \in B_\eta$, $\eta \leq \delta_0$, the iterates $P^k(z)$ stay in B_η and tend to y^* , the return

times $\tau_{\text{loc}}(P^k(z))$ lie in $[T/2, 2T]$, and between the k -th and the next return the orbit stays within $L|P^k(z) - y^*|_* \leq L\kappa^k\eta$ of Γ . Given ε , choose $\eta \leq \delta_0$ with $L\eta < \varepsilon$ and then δ as in Step 2: every x with $\text{dist}(x, \Gamma) < \delta$ stays within ε of Γ for all $t \geq 0$, and its distance to Γ tends to 0. $\square$

Lemma 5.13 (proved; saddle type). *Under the hypotheses of Lemma 5.12, except that $DP(y^*)$ has a simple real eigenvalue $\mu_1 > 1$ and its other eigenvalues have modulus less than 1, Γ is not orbitally stable.*

Proof. Choose coordinates $w = (\xi, \zeta) \in \mathbb{R} \times \mathbb{R}^2$ on Σ centred at y^* in which $DP(y^*) = \text{diag}(\mu_1, B)$, and a norm $|\cdot|$ on $\mathbb{R}^2$ with $|B| \leq \beta < 1$. Fix $\varepsilon > 0$ with $\beta + \varepsilon \leq \mu_1 - \varepsilon$ and $\mu_1 - \varepsilon > 1$, and $\delta_0 > 0$ such that $P(w) = DP(y^*)w + r(w)$ with $\|r(w)\| \leq \varepsilon\|w\|$ for $\|w\| = \max(|\xi|, |\zeta|) \leq \delta_0$. On the cone $\mathcal{K} = \{|\zeta| \leq |\xi|\}$, $w' = P(w)$ satisfies $|\xi'| \geq (\mu_1 - \varepsilon)|\xi|$ and $|\zeta'| \leq \beta|\zeta| + \varepsilon|\xi| \leq |\xi'|$, so $w' \in \mathcal{K}$ and $\|w'\| \geq (\mu_1 - \varepsilon)\|w\|$. Hence every $w \in \mathcal{K} \setminus \{0\}$, however close to 0, has an iterate with $\|P^k(w)\| > \delta_0$ whose predecessor has norm at most δ_0 ; by continuity of P , $\|P^k(w)\| \leq \Lambda\delta_0$ for a constant Λ . Transversality gives $c_0 > 0$ and $\eta_0 > 0$ with $\text{dist}(z, \Gamma) \geq c_0\|z\|$ for $z \in \Sigma$ with $\|z\| \leq \eta_0$: for $|s| \leq s_0$ small, the point $\varphi_s(y^*)$ of Γ satisfies $|z - \varphi_s(y^*)| \geq |x_i(\varphi_s(y^*)) - c| \geq a|s|$ with $a = \frac{1}{2}F_i(y^*)$, and $|z - \varphi_s(y^*)| \geq |z - y^*| - b|s|$ with $b = \max|F|$ near Γ , so $|z - \varphi_s(y^*)| \geq \frac{a}{a+b}|z - y^*|$; the rest of Γ is at a positive distance from y^* , hence from z when z is close to y^* ; and the norms are equivalent. Choosing δ_0 with $\Lambda\delta_0 \leq \eta_0$, orbits that start arbitrarily close to Γ reach the distance $c_0\delta_0$ from it, and Γ is not orbitally stable. $\square$

6 Proofs of Theorems 4 and 5 and Corollary 6

The references are to `data/certify_bistability.txt`; the program works at 96 bits with Taylor order $p = 20$ and a remainder tolerance $1 \cdot 10^{-18}$ relative to the scales $(100, 1, 1, 1, 10)$ of (u, m, n, h, E_l) . Every decimal bound it prints in stages 3 to 7 is rounded outward from its ball, and all 334 of them are read back as exact rationals and compared with the balls (stage 7, first check). Theorems 4 and 5 are the program's Theorems C and B; its Theorem A appears in the proof of Corollary 6.

Proof of Theorem 5. Fix E_l as in the theorem. The program carries E_l as a ball $\mathcal{E}$ that contains it: for 10.613 and 10.599 the ball of the decimal string at 96 bits, of radius below 10^{-28} , and for E_l^* a ball of radius below 10^{-25} (stage 3); every statement holds for each E_l in $\mathcal{E}$. (a) The program computes a candidate $\bar{z}$ on Σ_{20} by floating-point Newton steps, integrates the set $\{\bar{z}\} \times \mathcal{E}$ to its first return (a C^0 run) and the set $Z \times \mathcal{E}$, with the box $Z = \bar{z} + X$ of half-widths at most about $1.21 \cdot 10^{-15}$ (C^1 run over Z ; the half-widths are those at $E_l = 10.613$), each by a chain of steps as in Lemma 5.5, and checks that the two runs integrated sets that contain $\{\bar{z}\} \times \mathcal{E}$ and $Z \times \mathcal{E}$ (“the C^0 run integrated a set containing ...”). It obtains enclosures of $P(\bar{z})$ and of DP over Z (Lemma 5.6), the latter with entries of radius about $1.10 \cdot 10^{-9}$, and verifies $K \subset \text{int } Z$ (stage 4, “Krawczyk: unique fixed point of the first-return map in Z ”). By Lemma 5.9, for every $E_l \in \mathcal{E}$ the map P has exactly one fixed point z^* in Z , and $z^* \in K$. By Lemma 5.7 its orbit Γ_s is periodic with minimal period $\tau(z^*)$, which lies in the enclosure of τ over Z (Table 1), and since $K \subset \text{int } Z$, near z^* the map P is the local return map at z^* . The Gershgorin discs of $\mathbf{DP}(Z)$ (Lemma 5.10) all lie in the open unit disc (“all nontrivial Floquet multipliers of the stable orbit lie in $|\mu| < 1$ ”), and the first is disjoint from the other two (“multiplier discs ... are isolated”), so it contains exactly one multiplier μ_1 and the other two lie in the union of the others; Lemma 5.8 identifies these eigenvalues with the nontrivial multipliers, and Lemma 5.12 gives orbital asymptotic stability. The run passes Σ_{20} downward once and certifies the first return in 303 steps; its step enclosures show that every orbit from Z reaches $u \geq 95.953532$ (and the corresponding values at the other E_l).

(b) The same on Σ_5 , with a box of half-widths from about $2.03 \cdot 10^{-15}$ to $2.13 \cdot 10^{-14}$: Lemma 5.9 gives the fixed point, in $K \subset \text{int } Z$, and Lemma 5.7 the period and the local return map; the three discs are pairwise disjoint, the first is centred on the positive real axis and lies outside the closed unit disc, and the other two lie inside the open unit disc (“exactly one Floquet multiplier of the second orbit is real and > 1 ”). By Lemma 5.10(b), (c), μ_1 is real and larger than 1, and Γ_u is of saddle type; Lemma 5.13 shows that it is not orbitally stable. The step enclosures bound $u \leq 11.045813$ along every orbit from the box.

(c) Γ_s reaches $u \geq 95.953532$ and Γ_u stays below 11.045813 (at $E_l = 10.613$; similarly at the other values), so they are different (“the orbits are distinct”); neither is an equilibrium, because $\dot{u} > 0$ at their section points. Uniqueness holds in the Krawczyk boxes Z , whose largest half-widths are about $1.21 \cdot 10^{-15}$, $1.22 \cdot 10^{-15}$ and $1.22 \cdot 10^{-15}$ for Γ_s and $2.13 \cdot 10^{-14}$, $1.08 \cdot 10^{-14}$ and $2.13 \cdot 10^{-14}$ for Γ_u at the three values, and whose smallest half-widths for Γ_u are about $2.03 \cdot 10^{-15}$, $1.00 \cdot 10^{-15}$ and $1.94 \cdot 10^{-15}$ (stage 4, “C¹ run over Z (radius ...)”, which prints the three half-widths; `data/identify_stable_orbit.txt` prints the centres of the boxes of Γ_s exactly, and their half-widths, which are the printed floating-point radii rounded up to Arb’s radius format, exactly as well). The fixed points lie in the sets K (Lemma 5.9), and Table 3 lists outward roundings of K , which are wider than Z . $\square$

Proof of Theorem 4. Stage 4b cuts $[10.59, 10.62]$ into 60 intervals of width $5 \cdot 10^{-4}$ and checks that they cover it without gaps (“the pieces cover $[10.59, 10.62]$ ”). For each piece $\mathcal{E}$ it chooses a box Z on Σ_{20} , with half-widths at most about $4.6 \cdot 10^{-7}$, around an approximate fixed point, and runs two first-return computations over $Z \times \mathcal{E}$ as in Lemma 5.5: a C^0 run that shows $P_E(Z) \subset \text{int } Z$ for every $E \in \mathcal{E}$, and a C^1 run that bounds $\|DP_E(z)\|_\infty$ for all $(z, E) \in Z \times \mathcal{E}$, by the maximum over the rows of the sums of the moduli of the entries of the enclosure; it also checks that both runs integrated sets that contain $Z \times \mathcal{E}$. Every piece passes with a bound between 0.5413 and 0.5446 (“every piece: $P_E(Z)$ in $\text{int } Z$, both runs integrated a set containing $Z \times \text{piece}$, and $\sup \|DP_E\|_\infty < 1$ ”; each line of a piece prints the three results). By Lemma 5.11, P_E has exactly one fixed point in Z , and every eigenvalue of DP_E there has modulus at most 0.5446; the fixed point lies in $P_E(Z) \subset \text{int } Z$, so by Lemma 5.7 its orbit is periodic with minimal period in the enclosure of τ over $Z \times \mathcal{E}$ and near it P_E is the local return map, and by Lemma 5.12 the orbit is orbitally asymptotically stable. $\Gamma(E_l)$ denotes this orbit. When E_l is the common end of two pieces, each of the two boxes holds exactly one fixed point of P_{E_l} ; we take $\Gamma(E_l)$ from the piece on the left (at 10.613 and 10.599 the two fixed points are the same point, by Corollary 7). Each piece is carried as the ball $[lo, hi]$ of Arb, which contains the decimal interval. For the four pieces of Corollary 7 the boxes are recomputed, with the same printed lines, and printed exactly by `code/identify_stable_orbit.py`; for E_l in those pieces $\Gamma(E_l)$ is taken from the printed boxes. The union over the pieces of the period enclosures is $[16.005827509, 16.013912063]$, and the minimum over the pieces of the lower bounds of the maximum of u is 95.953 (stage 4b, last line, and stage 7). $\square$

Proof of Corollary 6. By Theorem 2(c), $J_{H1} \geq 9.773337 > 8$ for every $E_l \in [10.59, 10.62]$ (`data/certify_equilibria_hopf.txt`, part D), so by Theorems 1 and 2(a) the equilibrium at $J = 8$ is unique and locally asymptotically stable. Theorem 4 gives the orbitally asymptotically stable periodic orbit, Theorem 5(b) the orbit of saddle type at the three values, and Lemma 2.2 the equivalent statement in J . The program for Theorems 4 and 5 proves the equilibrium part again, independently, at 96 bits with the other implementation of Ψ (its Theorem A, stage 3): an interval Newton step on $J_{ss}(u) - 8$ over $U = [4.64089, 4.65911]$ with $J'_{ss}(U) \subset [2.3706, 2.4294]$ maps U into its interior; $J'_{ss} > 0$ on $W = [3.646624, 5.646624]$ (64 pieces); $J_{ss} - 8$ has no zero on $[-12, 115] \setminus W$ (70 interval evaluations) and none outside $[-12, 115]$ (the signs of Lemma 2.1); the equilibrium lies in the

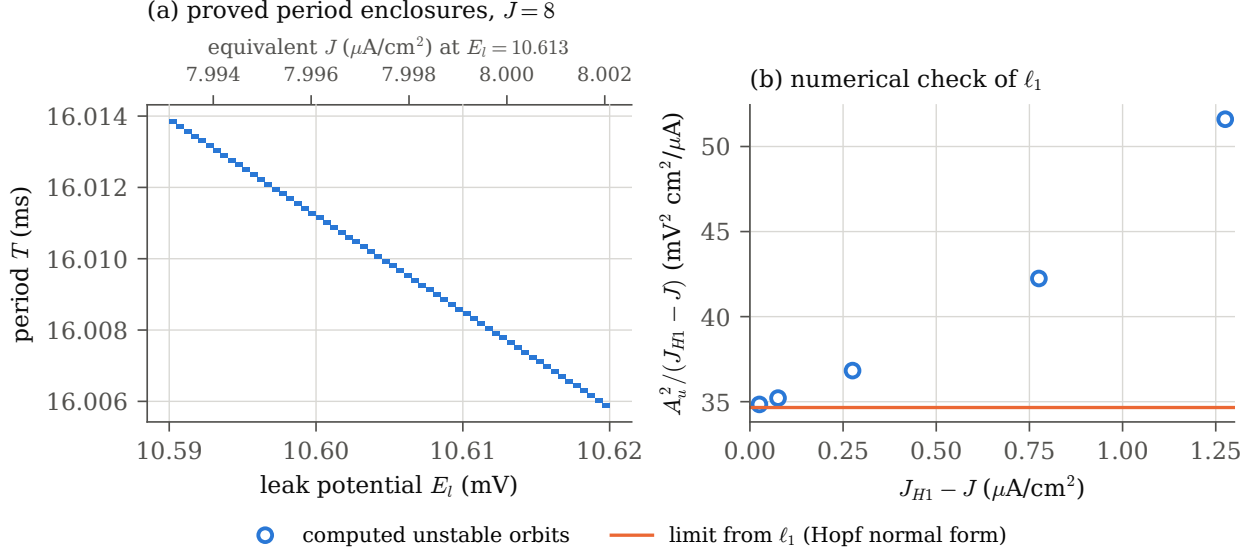


Figure 3: (a) Theorem 4: the proved enclosures of the period of $\Gamma(E_l)$ on each of the 60 pieces of the E_l interval; the top axis gives the equivalent current at $E_l = 10.613$ (Lemma 2.2). (b) Numerical: $A_u^2/(J_{H1} - J)$, where A_u is the peak-to-peak amplitude of u in mV, for the unstable periodic orbits computed in floating point at $J = 8.5$ to 9.75 ($E_l = 10.613$; Table 5), and its limit $16|q_u|^2 \operatorname{Re} \lambda'(J_{H1})/(\omega_1 \ell_1) \in [34.657379, 34.657380]$ predicted by the Hopf normal form from the proved values of Theorem 2 (Section 8).

box $u \in [4.64375, 4.65625]$, $m \in [0.0900353, 0.0901647]$, $n \in [0.3905195, 0.3906805]$, $h \in [0.42941, 0.43059]$; the enclosed characteristic polynomial has $a_1 \in [4.8848, 4.8952]$, $a_2 \in [1.2753, 1.3247]$, $a_3 \in [1.5908, 1.6092]$, $a_4 \in [0.1929, 0.2071]$, $D_2 \in [4.616, 4.784]$ and $D_3 \in [2.406, 3.594]$; and four pairwise disjoint Gershgorin discs, centred at -4.690211 , $-0.034513 \pm 0.566815i$ and -0.135017 with radii $6.9 \cdot 10^{-3}$, $8.4 \cdot 10^{-3}$ and $6.1 \cdot 10^{-5}$, lie in $\operatorname{Re} \lambda < 0$. $\square$

Proof of Corollary 7. The program `code/identify_stable_orbit.py` (output `data/identify_stable_orbit.txt`; 33 checks: 19 proof checks, 6 negative controls and 8 cross-checks; about 244 s with two worker processes) recomputes the fixed points of Theorem 5(a) as stage 4 of `code/certify_bistability.py` does, from the same floating-point candidate, and obtains the same sets K (cross-checks). With the function and the arguments of stage 4b (the box centres predicted from the stage-4 run at $E_l = 10.613$ and its derivative dz^*/dE_l) it recomputes the boxes of Theorem 4 on the pieces $[10.5985, 10.5990]$, $[10.5990, 10.5995]$, $[10.6125, 10.6130]$, $[10.6130, 10.6135]$, re-proves on each that $P_E(Z) \subset \operatorname{int} Z$ and $\sup \|DP_E\|_\infty < 1$ for every E_l in the piece, with runs that integrated sets containing $Z \times \text{piece}$, and prints the centres of the boxes and their half-widths (the floating-point radii rounded up to Arb's radius format) exactly; its lines for these pieces are those of stage 4b in `data/certify_bistability.txt` (cross-checks). For each value and each of these pieces that contains it, it checks that the ball of the value lies in the piece and that K lies in the interior of the box of the piece (the centre of K is at most 0.18 box radii from the centre of the box). The fixed point of Theorem 5(a) lies in K (Lemma 5.9), hence in the box, where P_E has exactly one fixed point (Lemma 5.11); so it is the point of $\Gamma(E_l)$. As negative controls, K at each value is tested against the box of a piece that does not contain the value, at least 10 box radii away, and against the box of its own piece with its half-widths shrunk to 0.1 of their values; both inclusions fail, so the test is not loose by a factor of ten. $\square$

7 The computer-assisted method

Ball arithmetic. All rigorous computations use Arb [22] through python-flint 0.9.0 (FLINT 3.6.0), in which a real number is a ball $[m \pm r]$ and a complex number a pair of balls. Every operation returns a ball that contains all results for all inputs in the input balls, and a comparison is true only if it holds for all points of the balls; an undecided comparison is false. The constants of (1) are exact rationals, converted to balls. Floating point (numpy, scipy, mpmath) proposes candidates, subdivision points, step sizes, the frames B of the Lohner sets, the Krawczyk preconditioner and the eigenvector matrices of the Gershgorin discs; a bad choice can make a check fail but not succeed wrongly. The three programs print every check and stop at the first failed check, or at an exception, with a nonzero exit status and, once they have started, a line in their report that says so.

Printed bounds. A bound printed as an interval $[a, b]$, or after $\leq$ or $\geq$, is rounded outward from its ball in exact rational arithmetic (`code/outward.py`), and each program re-reads all of its printed bounds as rationals and compares them with the balls (99, 334 and 12 bounds). A ball printed as $[m \pm r]$ is Arb’s own decimal enclosure. `code/hh_make_numbers.py` copies these bounds into the manuscript, or rounds them outward further, in exact rational arithmetic; no number of a theorem is typed by hand.

Checks. Each program counts its checks by kind (Table 4). A *proof* check is a step of a proof above. A *consistency* check verifies that an identity which a lemma proves exactly (such as $Aq = i\omega q$) holds within the enclosures. A *negative control* feeds a deliberately wrong input to the code of a certificate, which must refuse it: for example, the Hopf factorization test at $u_{H1} + 10^{-3}$, where $a_2 - \omega^2 - a_4/\omega^2 \in [-0.0004982, -0.0004981]$; mutated versions of (3); Krawczyk on boxes shifted by ten box radii or by 10^{-9} , and on the stable orbit’s box with the vector field at $J = 8.05$ (the centre of K lies $4.5 \cdot 10^{10}$ box radii away); Krawczyk with the enclosure of $DP(Z)$ widened by 0.5 in every entry, where the Newton point $\bar{z} - R(\mathcal{P}(\bar{z}) - \bar{z})$ still lies in Z , so that the test must see the term $(I - R(DP(Z) - I))X$; the check that the sets integrated contain $Z \times \mathcal{E}$, on a set a thousand times smaller than Z and on a set with E_l at the midpoint of its ball; an initial set on $u = 20$ with the crossing detected on $u = 19.999$; the equilibrium test at $J = 20$, where $D_3 \in [-18.86, -17.14]$; the contraction test on a box too small to contain the fixed points; on the usual box of a piece with the bound 1 replaced by 0.5, which the enclosure’s upper bound 0.5439 of $\sup \|DP_E\|_\infty$ does not meet (so that the verdict is shown to use the bound); and on the saddle orbit, where every member of the enclosure has $\|DP\|_\infty \geq 37.997$; the inclusion of K of Theorem 5(a) in the box of Theorem 4 of a piece that does not contain the value of E_l , and in its own box shrunk to 0.1; the integrator with the Taylor remainder set to 0, and the a priori test on boxes that do not contain the motion; and wrong vector fields, among them $g_{Na} = 120.0001$ against a high-precision reference solution. A *self-test* compares the code with a known answer (exact solutions of test equations, the Poincaré map of a planar system with known multipliers, mpmath references for the Hodgkin–Huxley flow in three regimes, the test systems for ℓ_1). *Cross-checks* (a non-rigorous recomputation, such as ℓ_1 from SymPy derivatives and mpmath eigenvectors, and $\text{Re } \lambda'(u_{Hi})$ and $J'_{ss}(u_{Hi})$ from central differences of mpmath eigenvalues, which agree with the enclosures to a relative $5.9 \cdot 10^{-27}$ and $2.7 \cdot 10^{-27}$; or the comparison of a recomputed line with the committed output of another program) and *numerical-only* checks enter no proof.

What an end-to-end run cannot show. A certificate can pass although a rigorous ingredient is missing, when the true solution satisfies the inequalities with a wide margin. The controls were

Table 4: The checks of the three programs by kind, from their outputs.

program	all	proof	consistency	controls	self-tests	cross-checks	numerical
<code>certify_equilibria_hopf.py</code>	40	17	6	5	8	4	–
<code>certify_bistability.py</code>	93	36	–	27	23	–	7
<code>identify_stable_orbit.py</code>	33	19	–	6	–	8	–

extended until the study below catches every mutation that is not marked weak. The program `code/mutation_study.py` makes deliberate mutations on copies of the programs, after running each unmutated program as a baseline (`data/mutation_study.txt`): of its 17 mutations, all 16 that are not marked weak stopped the program, 16 of them at a failed check and 0 by an exception; the mutation marked weak, which changes a term of the order of the floating-point error of a root, far below the enclosure radii, passed (1 of 1). The study is not part of any proof, and a caught mutation shows only that the checks see that particular error. The tests of (3) in `code/certify_equilibria_hopf.py` were added for this paper.

Trust base. The proofs assume that Arb and python-flint compute correct enclosures for the operations used (field operations, absolute values, complex conjugation, comparisons, `exp`, `sqrt`, the constant π , `hypgeom_1f1`, matrix products, `inv`, `solve` and `charpoly`), that FLINT’s exact rational Bernoulli numbers (`fmq.bernoulli`, used with Lemma 4.4(a)) are correct, that Python’s integers and fractions are exact, and that the programs do what Sections 4 to 6 describe.

8 Numerical observations

Nothing in this section is used in a proof. Where no other source is named, the numbers are from stage 2 (floating point, scipy DOP853 at relative tolerance 10^{-12} ; mpmath) and stage 6 of `data/certify_bistability.txt`; stage 6 runs the program’s Taylor integrator at 160 bits, whose steps carry rigorous remainders, inside a Newton iteration on midpoints that is not certified, so its values are approximations, not enclosures.

- *Rest and the equilibrium branch.* At $J = 0$ and $E_l = 10.613$ the equilibrium is at $u \approx 3.620669 \cdot 10^{-3}$ mV, not at 0 (Proposition 2.3). J_{ss} is increasing on a grid of $[-12, 250]$ with $J_{ss}(250) \approx 9418.0$, so numerically the equilibrium is unique up to about that current, beyond the range $[0, 4089.4815]$ of Remark 3.1.
- *The fold of cycles.* Along the branch of stable periodic orbits parametrized by the period, a multiplier crosses 1 at $J_{LPC} \approx 6.260321$ ($E_l = 10.613$) and 6.264545 ($E_l = E_l^*$), with period ≈ 19.8952 ms, so repetitive firing starts at about 50.3 Hz; the difference of the two currents is $0.3(10.613 - E_l^*)$, as Lemma 2.2 requires. Numerically the equilibrium and firing coexist for $J \in (J_{LPC}, J_{H1})$. This range is not simple: Guckenheimer and Oliva found chaos in it, near $J = 7.86$ [12]; we do not study it here (Section 11). The current $J = 8$ of Theorems 4 and 5 lies above these values.
- *The unstable orbits and the size of ℓ_1 .* Table 5 lists unstable periodic orbits computed between $J = 8.5$ and 9.75 . Near a Hopf point the normal form predicts that $(\text{peak-to-peak } u)^2 \approx 16|q_u|^2 \operatorname{Re} \lambda'(J_{H1})(J_{H1} - J)/(\omega_1 \ell_1)$, with q normalized as in (3); with the proved values of Theorem 2 and $|q_u| \in [0.9999042898, 0.9999042899]$ the factor is $[34.657379, 34.657380]$. The computed ratios decrease towards it as $J \rightarrow J_{H1}$, and a linear extrapolation through the

Table 5: Numerical: the unstable periodic orbits between $J = 8.5$ and J_{H1} at $E_l = 10.613$ (stage 2 of `data/certify_bistability.txt`), and the ratio of Figure 3(b), whose predicted limit is $[34.657379, 34.657380]$.

J	period (ms)	peak-to-peak u (mV)	$J_{H1} - J$	ratio
8.50	12.6126	8.1128	1.2754	51.60
9.00	11.6921	5.7233	0.7754	42.24
9.50	11.0248	3.1849	0.2754	36.83
9.70	10.7986	1.6298	0.0754	35.21
9.75	10.7448	0.9414	0.0254	34.84

two orbits closest to J_{H1} gives 34.65 (Figure 3(b)). This checks the magnitude of ℓ_1 , not only its sign. The orbits were located at each J separately, by bisecting perturbations of the equilibrium between a return to rest and a spike, the method that located the candidate for Γ_u at $J = 8$; that they and Γ_u belong to the family born at J_{H1} is suggested by the decreasing amplitudes, not proved.

- *The small cycles below J_{H2}* (`code/numerics_h2.py`, `data/numerics_h2.txt`; floating point). Integrating from a perturbation of the unstable equilibrium at $J = J_{H2} - \delta$, $E_l = 10.613$, for $\delta = 1, 0.5$ and 0.25 , the solution approaches a small periodic orbit, which Newton’s method on the return map to $\{u = u^*, \dot{u} > 0\}$ refines; its nontrivial multipliers have modulus at most 0.987 , so it is numerically stable, as Corollary 3(b) says. The ratio (peak-to-peak u)²/ $(J_{H2} - J)$ is $14.42, 14.36$ and 14.32 , decreasing towards the normal-form value $[14.292859, 14.292860]$ of Theorem 2, and a linear extrapolation through the last two gives 14.29 . So the size of ℓ_1 at J_{H2} is consistent with the numerics as well.
- *High-precision values at $J = 8$, $E_l = 10.613$* . Newton’s method on midpoints gives

$$T_s \approx 16.00771280383757999237802\dots, \quad \mu_1(\Gamma_s) \approx 0.07089678069271872876\dots,$$

$$T_u \approx 14.34360094817270692874908\dots,$$

and the multipliers $10.3028828212211696095\dots$ and $0.19683222913631744833\dots$ of Γ_u , inside the enclosures of Tables 1 and 2. They are the starting points of Figure 2.

9 Reproducibility

The programs need Python 3.11 with the packages of `code/requirements.txt` (python-flint 0.9.0, mpmath, SymPy, numpy, SciPy; matplotlib for the figures). From the folder of this paper:

command	what it proves, and time
<code>python3 code/certify_equilibria_hopf.py</code>	Proposition 2.3, Theorems 1, 2; 40 checks; about 7 s
<code>python3 code/certify_bistability.py</code>	Theorems 4, 5, Corollary 6; 93 checks; 53.1 min, of which stage 4b took 1707 s with 2 worker processes
<code>python3 code/identify_stable_orbit.py</code>	Corollary 7; 33 checks; about 244 s with two worker processes
<code>python3 code/numerics_h2.py</code>	the small cycles below J_{H2} (numerical, Section 8); a few minutes
<code>python3 code/mutation_study.py</code> <code>--workers 2</code>	the mutation study of Section 7 (not a proof); about 37 min
<code>python3 code/hh_make_numbers.py</code>	the numbers and tables of this paper, from <code>data/</code>
<code>python3 code/hh_make_figures.py</code>	the figures, about 10 s

`code/certify_bistability.py` accepts `--no-ball` (skip stage 4b, and so Theorem 4), `--quick` (skip the mpmath tests and stage 6), `--workers N` (the number of worker processes of stage 4b, default 2), `--resume` (read back the pieces of stage 4b that a checkpoint file holds for the same arguments and the same code, identified by the SHA-256 of the files that compute a piece, instead of recomputing them; the output says how many) and a debugging switch `--reuse-numeric`s, which reloads the floating-point candidates of stage 2 from an untracked file and says so in its output. Stage 4b appends each piece to the checkpoint as it completes and prints a line to the terminal, not to the report, which lists the pieces in order at the end.

Runs. The programs of this folder were run for this paper on 2026-09-27, as published in the companion repository, on a machine shared with other jobs and with at most two worker processes: `code/certify_equilibria_hopf.py` (40 checks), `code/certify_bistability.py` in full (93 checks in 53.1 min; stage 4b in 2 worker processes, with 0 pieces read from a checkpoint and the candidates of stage 2 computed in the run), `code/identify_stable_orbit.py` (33 checks), `code/numeric_h2.py` and `code/mutation_study.py`. Every check passed, and `data/` holds the outputs. The files of the proofs have the following SHA-256 prefixes:

file	SHA-256, first 16 digits
<code>code/certify_equilibria_hopf.py</code>	c2592f1e4559a8d4
<code>code/hh_ball.py</code>	754623075e174f8d
<code>code/certify_bistability.py</code>	49a0da01e7a03c56
<code>code/hh_arb.py</code>	88321fe660c541fd
<code>code/hh_lohner.py</code>	7dd88ec6057acff6
<code>code/certlib.py</code>	aab30999170cc70c
<code>code/ball_stable.py</code>	a7acfc9e5a168c49
<code>code/outward.py</code>	417e163289459752
<code>code/tests_integrator.py</code>	6e8417d0b6bff303
<code>code/testsys.py</code>	bbb47b4132b0164e
<code>code/shoot_float.py</code>	077dbbb104ba0f7f
<code>code/hh_numeric.py</code>	f48dc49e46fca474
<code>code/identify_stable_orbit.py</code>	83564add9d2465b4
<code>code/mutation_study.py</code>	782f5b11f93770cd
<code>data/certify_equilibria_hopf.txt</code>	add0ef88d9f2306f
<code>data/certify_bistability.txt</code>	62908b21151b309f
<code>data/identify_stable_orbit.txt</code>	3cc8546e0a84f4e9
<code>data/mutation_study.txt</code>	d472c4d9e775d01d

10 Limitations and what has been checked

What is not proved. The Hopf statements (Corollary 3) are local and give no size of the neighbourhoods. The periodic orbits are proved only at $J = 8$ and the currents equivalent to it; the saddle-type orbit only at three values of E_l , because the C^1 enclosures over boxes of the size the E_l interval requires are too wide for it. Uniqueness of the orbits holds only within their boxes, and the stable orbits of Theorems 4 and 5 are identified only at the three values of Corollary 7. Nothing is proved about other attractors, the basins, the separatrix, the fold of cycles or the global organization of the periodic orbits (Remark 3.2), or the equilibria for $J \geq 4089.4815$.

Sources of the proofs. Every source that a step of a proof uses was read in full where that step depends on it. The model is that of [19]: its summary of the equations and parameters (Part III, pp. 518–520, with eqs. (7), (15), (16), (26), the rate functions (12), (13), (20), (21), (23), (24), Table 3 and its footnote, the units and the temperature), and the pages where the rate functions

are introduced (pp. 510, 515, 516), read in the scanned pages at PubMed Central. Theorem H and formula (3) are used as stated in Kuznetsov’s encyclopedia article [26], read in full (revision 90964); the article cites his book [25] for the formula, which we have not checked against the book, and no proof here uses the book. The formula is also tested on systems with independently known ℓ_1 (Section 4), and its magnitude is consistent with the numerics at both Hopf points (Section 8). Theorem K is Rump’s Theorem 13.3 as he states it [34]; his Section 13, with Theorems 13.1 to 13.3 and their proofs, and his Lemma 10.5, which the proof of Theorem 13.3 uses, were read in full (in the version on the author’s web page). The facts on Bernoulli numbers and Kummer’s function are DLMF eqs. 13.2.2, 24.2.1, 24.2.2, 25.6.1, 25.6.2 and Table 24.2.1 [6], read. Gershgorin’s paper [10] was not read; Lemma 5.10 is proved here. Lohner [30] and Zgliczyński [39] are cited for the representation of sets in Section 5, which proves what it uses; we have read the abstract of [39], and [30] is a credit only. Krawczyk [24] and Moore [32] are credited for the test that Theorem K states, and were not read. Johansson’s papers [22, 23], of which we read the abstracts, describe the software, whose correctness is part of the trust base (Section 7).

Other sources, and the searches. Of the literature on the model we read in full [38, 12, 11, 17] and Chapter IV of [27]; in part [37, 18, 13, 8, 9, 31]; the abstract and the zbMATH review of [7], whose full text is available from its publisher by subscription only; abstracts or reviews of [36, 2, 14, 28, 29, 16, 35]; and [15, 33] from the papers that cite them. Of the computer-assisted work on other models we read the abstracts of [1, 5, 4, 3] (of [1] also the start of its first section). The searches for earlier work, from 2026-09-25 to 2026-09-27, were: zbMATH Open (the Hodgkin–Huxley model with Hopf, bistability, interval, computer, rigorous, proof, periodic existence, unique equilibria, stability, Lyapunov coefficient and subcritical; Hopf with interval arithmetic; the authors Hassard and Du), Semantic Scholar (computer-assisted proof, rigorous bistability, interval Hopf, and the papers citing [15] and [33]), Crossref, web searches, the arXiv (twenty-five searches on 2026-09-27, twenty-three of them in the abstracts, among them “Hodgkin–Huxley” with Hopf, bistability, computer-assisted, interval arithmetic, rigorous, periodic orbit, limit cycle, Lyapunov coefficient, validated, proof and subcritical, and “computer-assisted proof” with neuron and with Hopf bifurcation) and PubMed (four searches on 2026-09-27 for the Hodgkin–Huxley model with computer-assisted, interval or rigorous methods, with bistability and a proof, with a subcritical Hopf point or a Lyapunov coefficient, and with the existence of periodic solutions). None found a proof of the results of this paper for the space-clamped equations at the 1952 constants; the hits closest to the subject treat other models, modified or stochastic versions of the equations, or numerics. The statements of priority in Section 1 are limited to these searches and to what was read. In particular, whether the location and the criticality of the two Hopf points, or statements about the branch of equilibria, have been proved before is not settled: Du and Hassard may contain them.

What has been checked. The proofs, the programs and the claims were checked within the project, in six adversarial readings by separate AI agent sessions and by the mutation study of Section 7; none of these is an outside review. Every finding that would make a stated result false was fixed, and the records of the readings are in the folder `notes/` of the companion repository.

11 Future work

Guckenheimer and Oliva’s chaotic dynamics near $J = 7.86$ [12] is the natural next target. An unpublished numerical study of ours, not part of this paper, relocates their periodic points as fixed

points of the return map to $u = 4.5$ crossed with u increasing, and finds two h-sets around two orbits of the branch born at J_{H1} whose covering relations hold in dense sampling, avoiding the multiplier of order 10^7 of one of their orbits; a proof would need of the order of 10^3 to 10^4 box integrations with the C^1 method of Section 5. Other open steps are a proof of the fold of cycles, which would bound the bistable range from below, the saddle-type orbit over the whole E_I interval, and statements about the boundary between the basins. The propagated action potential of the Hodgkin–Huxley cable equation at the 1952 constants is a separate problem, for which the literature we reached has proofs only for modified systems.

Data availability. The programs of this paper, their outputs and the scripts that write the numbers, tables and figures of the manuscript are published with this paper as its companion repository, <https://github.com/ChaseHendrick/hh-dynamics>, whose releases are archived on Zenodo under the concept DOI <https://doi.org/10.5281/zenodo.22996259>: the programs in `code/` (with `requirements.txt`) and their outputs in `data/`.

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